

"Volume fraction estimation:"

```
N[1.660539040 * 10^(-27) * 1000 * 300 *
  1000 / ((2 * Pi * (11 * 10^(-7) / 2)^2 * 5.5 * 10^(-7))) / 1.25]
N[1.660539040 * 10^(-27) * 1000 * 650 * 2 / (4 / 3 * Pi * 10^(-7 * 3)) / 1.25]
N[0.30 * 0.38]
N[Sqrt[0.3 * (1 - 0.3)] * 1.25 * 0.4 * 0.17]
```

Volume fraction estimation:

0.381235

0.412281

0.114

0.0389519

"Checking that $1/lc^{(4-D)}$ size distribution gives a power-law ACF $\sim 1/r^{(3-D)}$:"

```
Integrate[Exp[-r/lc]/lc^(4-d), {lc, lmin, lmax}]
```

```
ConditionalExpression[
```

$$l_{\max}^{-3+d} \text{ExpIntegralE}\left[-2+d, \frac{r}{l_{\max}}\right] - l_{\min}^{-3+d} \text{ExpIntegralE}\left[-2+d, \frac{r}{l_{\min}}\right],$$

$$\left(\left(\text{Im}[l_{\max}] \geq \text{Im}[l_{\min}] \ \&\& \ \text{Im}[l_{\min}] \text{Re}[l_{\max}] \leq \text{Im}[l_{\max}] \text{Re}[l_{\min}] \right) \ || \right.$$

$$\left. \left(\text{Im}[l_{\min}] \text{Re}[l_{\max}] \geq \text{Im}[l_{\max}] \text{Re}[l_{\min}] \ \&\& \ \text{Im}[l_{\max}] \leq \text{Im}[l_{\min}] \right) \right) \ \&\&$$

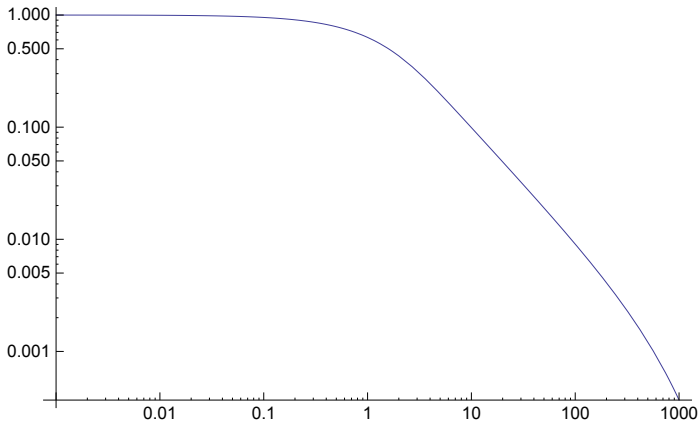
$$\left(\left(\text{Re}\left[\frac{l_{\min}}{l_{\max} - l_{\min}}\right] \geq 0 \ \&\& \ l_{\max} l_{\min} \neq l_{\min}^2 \right) \ || \ \frac{l_{\min}}{l_{\max} - l_{\min}} \notin \text{Reals} \ || \ \text{Re}\left[\frac{l_{\min}}{l_{\max} - l_{\min}}\right] \leq -1 \right)]$$

```
corf[r_, d_, lmin_, lmax_] := d * lmin^(3-d) *
```

$$\left(l_{\max}^{-3+d} \text{ExpIntegralE}\left[-2+d, \frac{r}{l_{\max}}\right] - l_{\min}^{-3+d} \text{ExpIntegralE}\left[-2+d, \frac{r}{l_{\min}}\right] \right) /$$

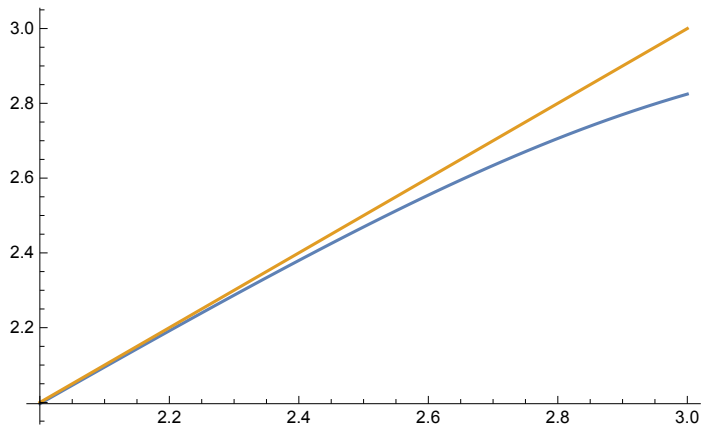
$$(d / (3-d) * (1 - (l_{\min}/l_{\max})^{(3-d)}));$$

```
LogLogPlot[{corf[r, 2, 1, 1000]}, {r, 0.001, 1000}, PlotRange -> All]
```

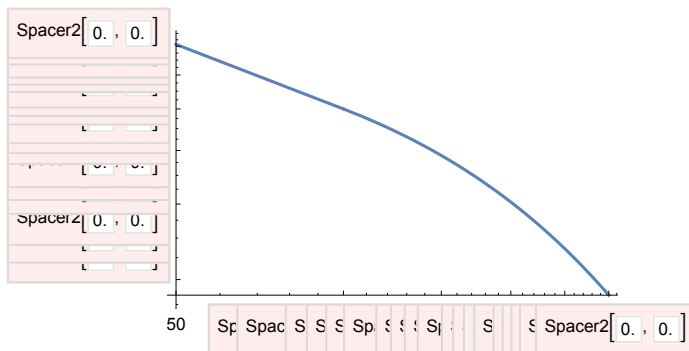


$$\tilde{\Sigma}^2 = \frac{2R\sigma_{n\Delta}^2}{\pi} \frac{k_c L x^3 N A^2}{(1 + x^2(4 + N A^2))(1 + 4x^2)} + \frac{R\sigma_{n\Delta}^2}{4} \left(1 - \frac{1}{\sqrt{1 + (xNA)^2}} \right)$$

```
mscale[d_, lmin_, lmax_] := NIntegrate[
  (3 - d) / (1 - (lmin / lmax)^(3 - d)) * (lmin / lc)^(4 - d) / lmin * lc^3, {lc, lmin, lmax}];
r1 = 300;
power[d_] :=
  (Log[mscale[d, 1, r1]] - Log[mscale[d, 1, r1 - 0.1]]) / (Log[r1] - Log[r1 - 0.1]);
LogLogPlot[{mscale[4, 1, r], r^4}, {r, 50, 400}];
Plot[{power[d], d}, {d, 2, 3}]
```



```
d1 = 2.5;
b1 = 100;
LogLogPlot[{1/x^(3 - d1) * UnitStep[b1 - x] + UnitStep[x - b1] *
  1/b1^(3 - d1) * Exp[-(x - b1) * (3 - d1) / b1]}, {x, 50, 300}, PlotRange -> All]
```



```

InverseLaplaceTransform[
  1/x^(3-d)*UnitStep[b-x]+UnitStep[x-b]*1/b^(3-d)*Exp[-(x-b)*(3-d)/b], x, s]
{ Indeterminate          b == x
  -b^(-3+d)*e^(b*s)*DiracDelta[-(3+d)/b + s]  b <= x
  -((-3+d)*s^(2-d)/Gamma[4-d])              b >= x
  0                                           True
}

Integrate[(3-d)^2*Gamma[3-d]*(rmin/s)^(4-d), {s, rmin, Infinity}]
Gamma[4-d]*rmin

ConditionalExpression[1, Re[d] < 3 && Re[rmin] > 0 && rmin == Re[rmin]]

size[lc_, d_, lmin_, lmax_] := (3-d)/(1-(lmin/lmax)^(3-d))*(lmin/lc)^(4-d)/lmin;
surfaceterm = 0;
massfixed = 1;
nachoise = 0;
dexpand = 2.5;

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = naistorm;
phi = 0.32;
riinc = 0.1899*0 + (0.1899*1/2 + 0.17*1/2 + 0.185*0);
sn = N[1/Sqrt[3]*(0.4-0.1)/2*0.17*1.25]*0 +
  Sqrt[Exp[4*(1-Sqrt[1+1/2*Log[0.13/0.45]])^2]-1]*0.13*0.17*1.25*0 +
  Sqrt[phi*(1-phi)]*0.38*riinc*1.25
ric = 1.332 + riinc*1.25*0.38*phi;
rig = 1.517 + 1.53*0;
rim = 1.337;
rcoeff =
  (4*ric*rig/(rig+ric)^2*(rig-ric)/(rig+ric)*(rig+rim)^2/(rig-rim)^2)^2;
kt[l_] := 2*Pi/lambda*ric*l*1000;
xp[lc_] := 2*Pi/lambda*ric*lc;
na = nachoise*Min[(1.4+nai)/rig, 1] + (1-nachoise)*1.4/rig;
s2[lc_, y_] :=
  2/Pi*(kt[y]*xp[lc]^3*na^2)/(1+xp[lc]^2*(4+na^2))/(1+xp[lc]^2*4) +
  surfaceterm*1/4*(1-1/Sqrt[1+(xp[lc]*na)^2]);
sigma[d_, lmin_, lmax_, y_] := Sqrt[NIntegrate[(3-d)/(1-(lmin/lmax)^(3-d))*
  (lmin/lc)^(4-d)/lmin*s2[lc, y], {lc, lmin, lmax}]];
sigmanorm[d_, lmin_, lmax_, t_] := sigma[d, lmin, lmax, t]*2*Pi/(na*Sqrt[kt[t]])^1;
sigmaexperiment[d_, lmin_, lmax_, t_] := sigma[d, lmin, lmax, t]*Sqrt[rcoeff]*sn;
sigmathinfilm = Sqrt[2]*rcoeff^0.5*(ric-rim)/(ric+rim);
sigmaexperimentwthinfilm[d_, lmin_, lmax_, t_] :=
  Sqrt[sigmaexperiment[d, lmin, lmax, t]^2 + sigmathinfilm^2];

mass[d_] := 500000*massfixed + (1-massfixed)*4*10^(-7)*Exp[11.1*d];
thick = 1.75*0 + 1*0 + 1.75/2*0 + lambda/1000*rig/2/nai^2;
"Linear approximation for Sigma_norm for 2<D<3"

```

```

a = sigmanorm[2.5, 1, mass[2.5]^(1/2.5), thick];
b = 1/D[sigmanorm[d, 1, mass[d]^(1/d), thick], d] /. d -> 2.5
c = N[(-a + 1/b * 2.5) * b]
"Linear approximation for Sigma_experiment at D=2.2"
a3 = sigmaexperiment[2.2, 1, mass[2.2]^(1/2.2), thick];
b3 = 1/D[sigmaexperiment[d, 1, mass[d]^(1/d), thick], d] /. d -> 2.2
c3 = N[(-a3 + 1/b3 * 2.2) * b3]
"Linear approximation for Sigma_experiment at D=D_expand"
Text["D_expand=", dexpand]
a4 = sigmaexperiment[dexpand, 1, mass[dexpand]^(1/dexpand), thick];
b4 = 1/D[sigmaexperiment[d, 1, mass[d]^(1/d), thick], d] /. d -> dexpand
c4 = N[(-a4 + 1/b4 * dexpand) * b4]
df[d_] := If[d <= 3, d, 3];
Show[Plot[{sigmaexperiment[d, 1, mass[df[d]]^(1/df[d]), thick],
  0, 0 * sigmaexperimentwthinfilm[d, 1, mass[df[d]]^(1/df[d]), thick]}],
  {d, 1, 3}, PlotRange -> All], Plot[{1/b4 * (d - c4)}, {d, 2, 3}, PlotRange -> All]]
Show[Plot[{sigmanorm[d, 1, mass[df[d]]^(1/df[d]), thick], 0}, {d, 1, 3}, PlotRange -> All],
  Plot[{1/b * (d - c)}, {d, 2, 3}, PlotRange -> All]]
a2 = sigmanorm[2.5, 1, mass[2.5]^(1/2.5), thick];
b2 = 1/D[sigmanorm[d, 1, mass[2.5]^(1/2.5), thick], d] /. d -> 2.5;
c2 = N[(-a2 + 1/b2 * 2.5) * b2];
Show[Plot[{sigmanorm[d, 1, mass[2.5]^(1/2.5), thick], 0,
  0 * sigmanorm[d, 1, (0.1 * mass[d])^(1/2.5), thick]}, {d, 0.5, 4.5}, PlotRange -> All],
  Plot[{1/b2 * (d - c2)}, {d, 2, 3.5}, PlotRange -> All]]
0.0398726

```

Linear approximation for Sigma_norm for 2<D<3

```

NIntegrate: Ic = 500000.1/2 is not a valid limit of integration.
NIntegrate: Ic = 500000.1/2 is not a valid limit of integration.
NIntegrate: Ic = 500000.1/2 is not a valid limit of integration.
General: Further output of NIntegrate::nlim will be suppressed during this calculation.

```

2.84127

1.47292

Linear approximation for Sigma_experiment at D=2.2

```

NIntegrate: Ic = 500000.1/2 is not a valid limit of integration.
NIntegrate: Ic = 500000.1/2 is not a valid limit of integration.
NIntegrate: Ic = 500000.1/2 is not a valid limit of integration.
General: Further output of NIntegrate::nlim will be suppressed during this calculation.

```

6.32748

1.35662

Linear approximation for Sigma_experiment at D=D_expand

Text [D_expand=, 2.5]

NIntegrate: $lc = 500000. \frac{1}{d}$ is not a valid limit of integration.

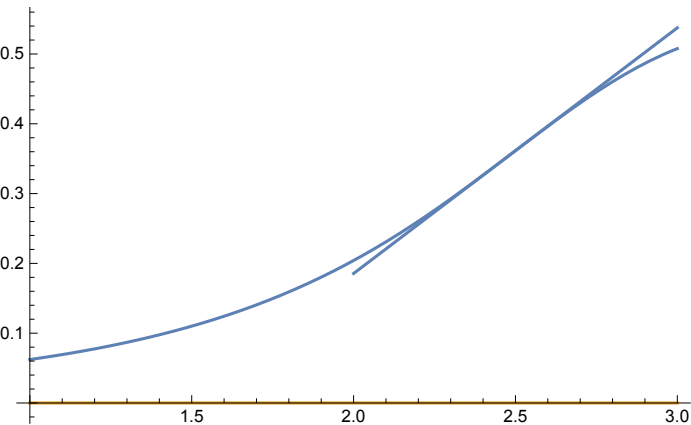
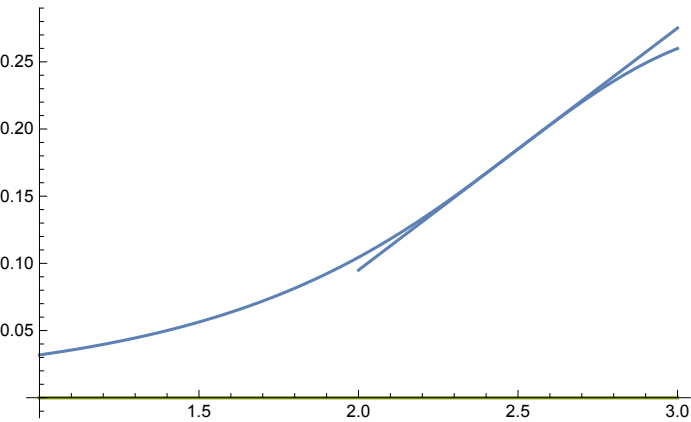
NIntegrate: $lc = 500000. \frac{1}{d}$ is not a valid limit of integration.

NIntegrate: $lc = 500000. \frac{1}{d}$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

5.55008

1.47292

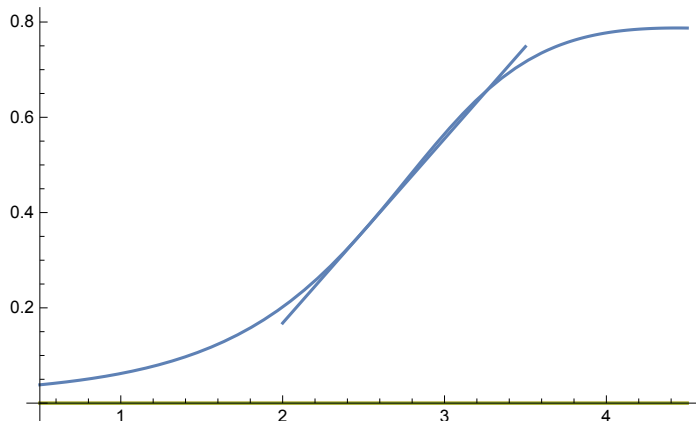


NIntegrate: The integrand $\frac{0.0000690308 (3 - d) \left(\frac{1}{lc}\right)^{1-d}}{(1 - 0.00525306^{3-d}) (1 + 0.000858513 lc^2) (1 + 0.00104131 lc^2)}$ has evaluated to non-numerical values for all sampling points in the region with boundaries {{1, 190.365}}.

NIntegrate: The integrand $\frac{0.0000690308 (3 - d) \left(\frac{1}{lc}\right)^{1-d}}{(1 - 0.00525306^{3-d}) (1 + 0.000858513 lc^2) (1 + 0.00104131 lc^2)}$ has evaluated to non-numerical values for all sampling points in the region with boundaries {{1, 190.365}}.

NIntegrate: The integrand $\frac{0.0000690308 (3 - d) \left(\frac{1}{lc}\right)^{1-d}}{(1 - 0.00525306^{3-d})(1 + 0.000858513 lc^2)(1 + 0.00104131 lc^2)}$ has evaluated to non-numerical values for all sampling points in the region with boundaries {{1, 190.365}}.

General: Further output of NIntegrate::inumr will be suppressed during this calculation.



⋮

```
Sqrt[0.051^2 - 0.009^2] * 3.04 * 6.75 + 1.515
Sqrt[0.051^2 - 0.009^2] * 3.04 * 7.63 + 1.473
Sqrt[0.051^2 - 0.009^2] * 3.04 * 7.23 + 1.51
Sqrt[0.051^2 - 0.009^2] * 3.04 * 9.39 + 1.19
```

2.5451

2.63739

2.61335

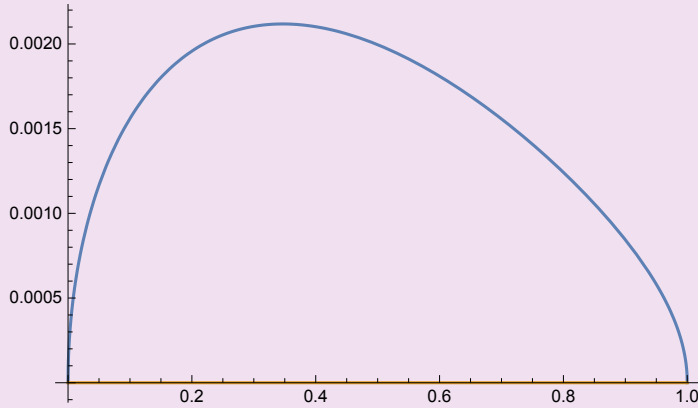
2.62298

Effect of chromatin density (very weak effect) :

```

ri[x_] := 1.334 + 0.18 * x * 1.25 * 0.45;
x0 = 0.1;
nn[x_] := ri[x + (1 - x) * x0];
ng = 1.517;
s[x_] := (ng - nn[x]) / (ng + nn[x]) * Sqrt[x * (1 - x)] * (ri[1] - ri[x0]);
Plot[{s[x], 0}, {x, 0., 1}]

```



```

surfaceterm = 0;
massfixed = 1;
nachoise = 0;

dexpand = 2.5;
mf = 1000000;
cvcdefault = 0.35;
phinucleoplasma = 0.1;

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = nailive;

rig = 1.517 + 1.53 * 0;
rim = 1.337;

riinc = 0.1899 * 0 + (0.1899 * 1 / 2 + 0.17 * 1 / 2 + 0.185 * 0);
purechromdensity = 0.555;
ri[x_] := 1.332 + riinc * purechromdensity * x;
sn[phi_] := Sqrt[phi * (1 - phi)] * (ri[1] - ri[phinucleoplasma]);

ric[phi_] := ri[phi + (1 - phi) * phinucleoplasma];
size[lc_, d_, lmin_, lmax_] :=
  (3 - d) / (1 - (lmin / lmax)^(3 - d)) * (lmin / lc)^(4 - d) / lmin;
rcoeff[phi_] := (2 * rig / (rig + ric[phi]) * 2 * ric[phi] / (rig + ric[phi]) *
  (rig - ric[phi]) / (rig + ric[phi]) * (rig + rim)^2 / (rig - rim)^2)^2;
kt[l_, phi_] := 2 * Pi / lambda * ric[phi] * 1 * 1000;

```

```

xp[lc_, phi_] := 2 * Pi / lambda * ric[phi] * lc;
na = nachoice * Min[(1.4 + nai) / rig, 1] + (1 - nachoice) * 1.4 / rig;
s2[lc_, y_, phi_] :=
  2 / Pi * (kt[y, phi] * xp[lc, phi]^3 * na^2) / (1 + xp[lc, phi]^2 * (4 + na^2)) /
    (1 + xp[lc, phi]^2 * 4) + surfaceterm * 1 / 4 * (1 - 1 / Sqrt[1 + (xp[lc, phi] * na)^2]);
sigma[d_, lmin_, lmax_, y_, phi_] := Sqrt[NIntegrate[(3 - d) / (1 - (lmin / lmax)^(3 - d)) *
  (lmin / lc)^(4 - d) / lmin * s2[lc, y, phi], {lc, lmin, lmax}]];
sigmanorm[d_, lmin_, lmax_, t_, phi_] := sigma[d, lmin, lmax, t, phi] *
  2 * Pi / (na * Sqrt[kt[t, phi]])^1;
sigmaexperiment[d_, lmin_, lmax_, t_, phi_] :=
  sigma[d, lmin, lmax, t, phi] * Sqrt[rcoeff[phi]] * sn[phi];
sigmathinfilmm[phi_] := Sqrt[2] * rcoeff[phi]^0.5 * (ric[phi] - rim) / (ric[phi] + rim);
sigmaexperimentwithinfilm[d_, lmin_, lmax_, t_, phi_] :=
  Sqrt[sigmaexperiment[d, lmin, lmax, t, phi]^2 + sigmathinfilmm[phi]^2];

mass[d_] := mf * massfixed + (1 - massfixed) * 4 * 10^(-7) * Exp[11.1 * d];
thick = lambda / 1000 * rig / 2 / nai^2;

"Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default"
Text["D_expand=", dexpand, "CVC default=", cvcdefault]
a4 = sigmaexperiment[dexpand, 1, mass[dexpand]^(1/dexpand), thick, cvcdefault];
b4 = 1 / D[sigmaexperiment[d, 1, mass[d]^(1/d), thick, cvcdefault], d] /. d -> dexpand
c4 = N[(-a4 + 1 / b4 * dexpand) * b4]
df[d_] := If[d <= 3, d, 3];
Show[Plot[{sigmaexperiment[d, 1, mass[df[d]]^(1/df[d]), thick, cvcdefault], 0},
  {d, 1, 3}, PlotRange -> All], Plot[{1 / b4 * (d - c4)}, {d, 2, 3}, PlotRange -> All]]

dalpha[alpha_] := 5 / 3 + 0.3 / (alpha - 2)^(2 / 3);
sigmaexperimentalalpha[alpha_, lmin_, lmax_, t_, phi_] :=
  sigmaexperiment[dalalpha[alpha], lmin, lmax, t, phi];
aexpand = 2.2;
Text["alpha_expand=", aexpand, "CVC default=", cvcdefault]
a4a = sigmaexperimentalalpha[aexpand, 1,
  mass[dalalpha[aexpand]]^(1/dalalpha[aexpand]), thick, cvcdefault];
b4a = 1 / D[sigmaexperimentalalpha[alpha, 1, mass[dalalpha[alpha]]^(1/dalalpha[alpha]),
  thick, cvcdefault], alpha] /. alpha -> aexpand
c4a = N[(-a4a + 1 / b4a * aexpand) * b4a]
df[d_] := If[d <= 3, d, 3];
Show[
  Plot[{sigmaexperimentalalpha[alpha, 1, mass[df[dalalpha[alpha]]]^(1/df[dalalpha[alpha]]),
    thick, cvcdefault], 0}, {alpha, 2.1, 2.35}, PlotRange -> All],
  Plot[{1 / b4a * (alpha - c4a)}, {alpha, 2.1, 2.35}, PlotRange -> All]]

Plot[{sigmaexperiment[dexpand, 1, mass[df[dexpand]]^(1/df[dexpand]), thick, phi], 0},
  {phi, 0.2, 0.5}, PlotRange -> All]
Plot3D[{sigmaexperiment[d, 1, z^(1/d), thick, cvcdefault]},
  {d, 2, 3}, {z, 100000, 1000000}, PlotRange -> All]
Plot3D[{sigmaexperiment[d, 1, mf^(1/d), thick, phi]},

```



```
{d, 2, 3}, {phi, 0.1, 0.6}, PlotRange -> All]
```

```
acoeff[z_, phi_] := sigmaexperiment[dexpand, 1, z^(1/dexpand), thick, phi];
bcoeff[z_, phi_] := 1/D[sigmaexperiment[d, 1, z^(1/d), thick, phi], d] /. d -> dexpand;
d0[z_, phi_] := (-acoeff[z, phi] + 1/bcoeff[z, phi] * dexpand) * bcoeff[z, phi];
Plot3D[{bcoeff[z, phi]}, {z, 100000, 1000000}, {phi, 0.2, 0.5}, PlotRange -> All]
Plot3D[{d0[z, phi]}, {z, 100000, 1000000}, {phi, 0.2, 0.5}, PlotRange -> All]
```

Linear approximation for Sigma_experiment at D=D_expand and phi=CVC_default

Text[D_expand=, 2.5, CVC default=, 0.35]

NIntegrate: $lc = 1. \times 10^{6 \frac{1}{d}}$ is not a valid limit of integration.

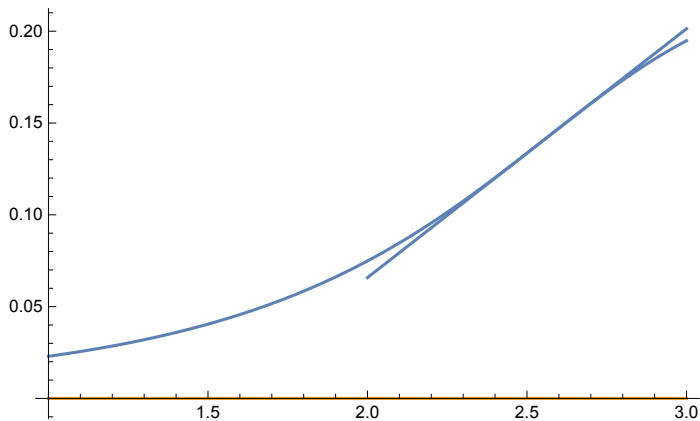
NIntegrate: $lc = 1. \times 10^{6 \frac{1}{d}}$ is not a valid limit of integration.

NIntegrate: $lc = 1. \times 10^{6 \frac{1}{d}}$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

7.37887

1.51432



Text[alpha_expand=, 2.2, CVC default=, 0.35]

NIntegrate: $lc = 1. \times 10^{6 \frac{1}{1.66667 + \frac{0.3}{(-2 + \alpha)^{2/3}}}}$ is not a valid limit of integration.

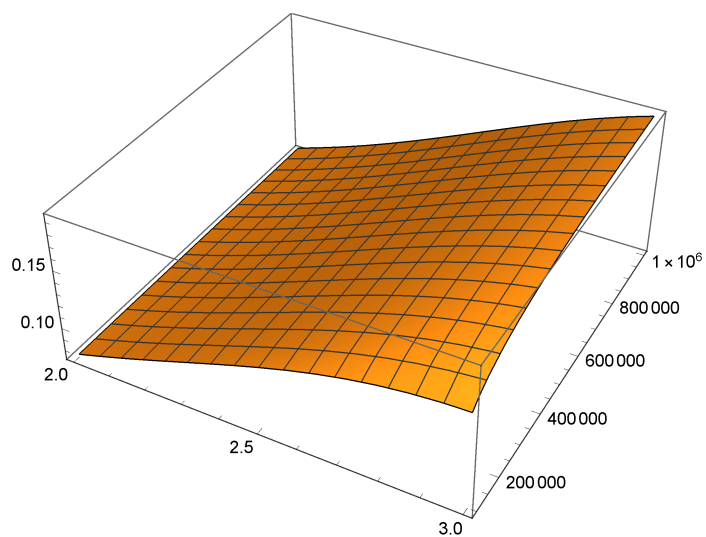
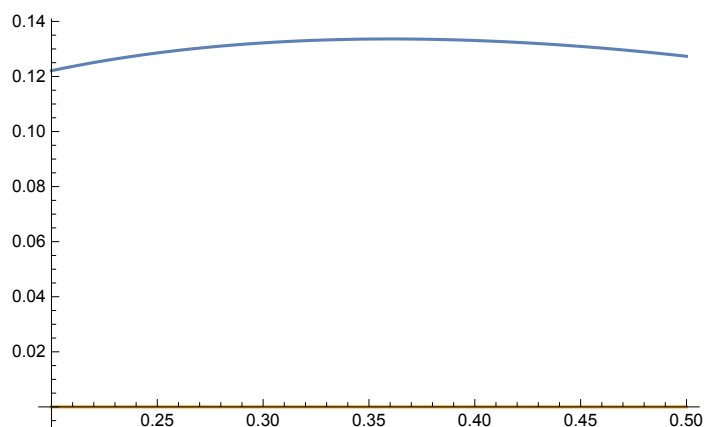
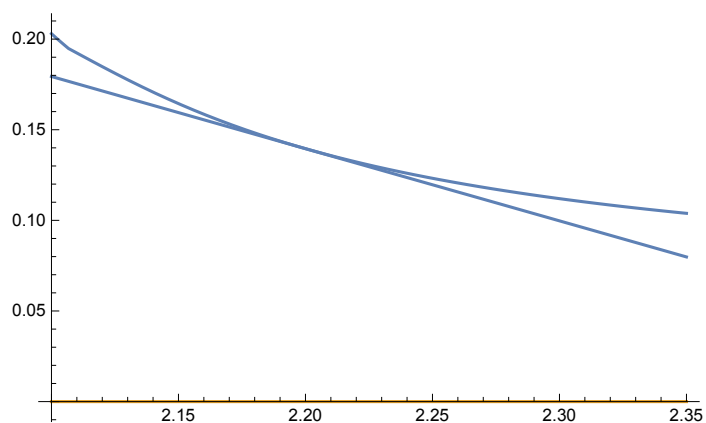
NIntegrate: $lc = 1. \times 10^{6 \frac{1}{1.66667 + \frac{0.3}{(-2 + \alpha)^{2/3}}}}$ is not a valid limit of integration.

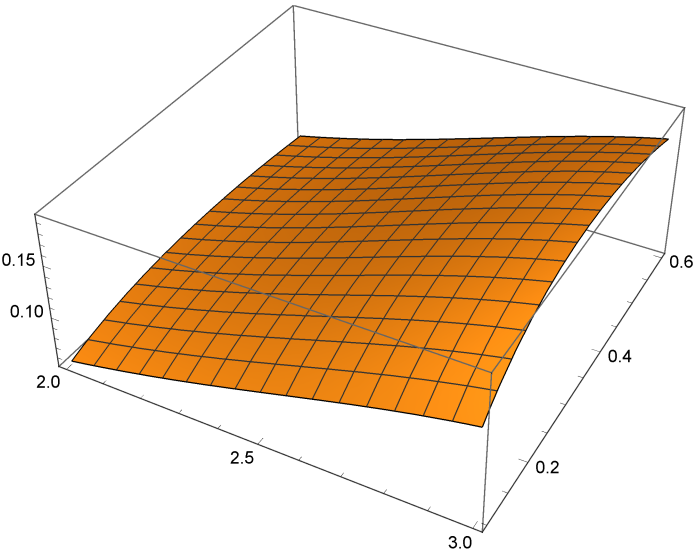
NIntegrate: $lc = 1. \times 10^{6 \frac{1}{1.66667 + \frac{0.3}{(-2 + \alpha)^{2/3}}}}$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

-2.51121

2.55042



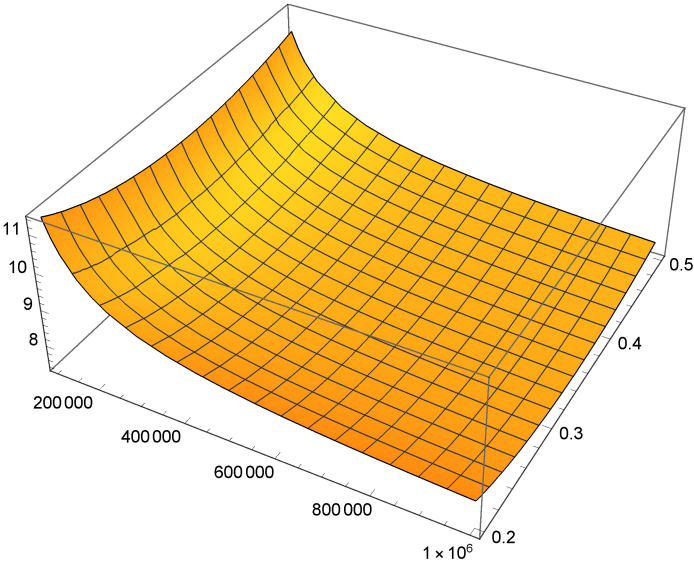


NIntegrate: $lc = 100064. \frac{1}{\pi}$ is not a valid limit of integration.

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NIntegrate: $lc = 100064. \frac{1}{\pi}$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

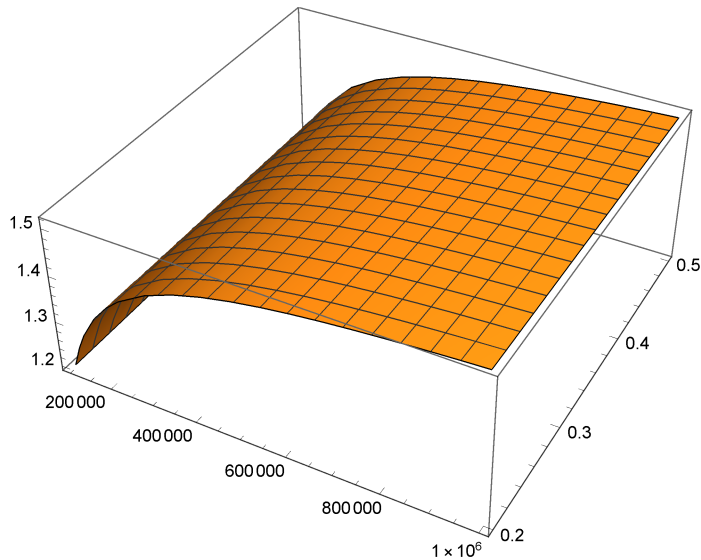


NIntegrate: $lc = 100064. \frac{1}{\pi}$ is not a valid limit of integration.

NIntegrate: $lc = 100064. \frac{1}{\pi}$ is not a valid limit of integration.

NIntegrate: $lc = 100064. \frac{1}{\pi}$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.



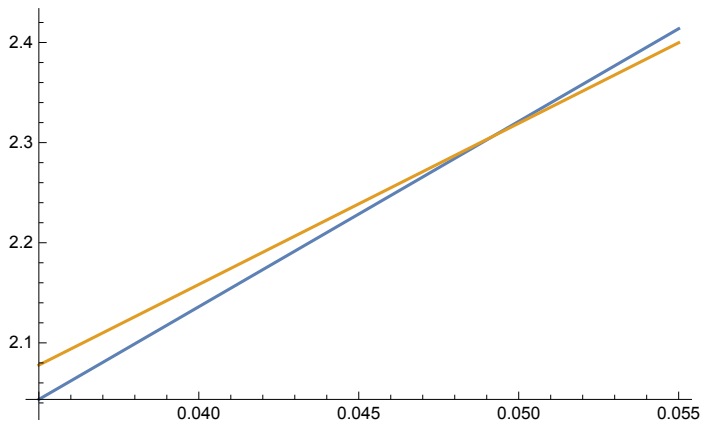
```
Plot[{x * 3 * bcoeff[250000, 0.35] + d0[250000, 0.35],
      x * 3 * bcoeff[1000000, 0.35] + d0[1000000, 0.35]}, {x, 0.035, 0.055}]
```

NIntegrate: $lc = 250000.\frac{1}{\pi}$ is not a valid limit of integration.

NIntegrate: $lc = 250000.\frac{1}{\pi}$ is not a valid limit of integration.

NIntegrate: $lc = 250000.\frac{1}{\pi}$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.



```
Integrate[2 / Pi * x^(d - 1) / (1 + (2 * k * x)^2)^2 +
          1 / 4 * x^(d - 4) * (1 - 1 / Sqrt[1 + (k * n * x)^2]), {x, 1, rm}]
```

$$\text{ConditionalExpression}\left[\frac{1}{4} \left(-\frac{1}{-3+d} - \frac{8}{d\pi} + \frac{32k^2}{(2+d)(1+4k^2)\pi} + \frac{16dk^2}{(2+d)(1+4k^2)\pi} + \frac{\text{Hypergeometric2F1}\left[-\frac{1}{2}, \frac{1}{2}(-3+d), \frac{1}{2}(-1+d), -k^2n^2\right]}{-3+d} \right), \right.$$

$$\begin{aligned}
& \frac{k^2 n^2 \operatorname{Hypergeometric2F1}\left[-\frac{1}{2}, \frac{1}{2}(-1+d), \frac{1+d}{2}, -k^2 n^2\right]}{-1+d} + \\
& \frac{k^4 n^4 \operatorname{Hypergeometric2F1}\left[\frac{1}{2}, \frac{1+d}{2}, \frac{3+d}{2}, -k^2 n^2\right]}{1+d} + \\
& \frac{32 k^2 \operatorname{Hypergeometric2F1}\left[1, 1+\frac{d}{2}, 2+\frac{d}{2}, -4 k^2\right]}{(2+d) \pi} - \\
& \frac{16 d k^2 \operatorname{Hypergeometric2F1}\left[1, 1+\frac{d}{2}, 2+\frac{d}{2}, -4 k^2\right]}{(2+d) (1+4 k^2) \pi} - \\
& \frac{64 d k^4 \operatorname{Hypergeometric2F1}\left[1, 1+\frac{d}{2}, 2+\frac{d}{2}, -4 k^2\right]}{(2+d) (1+4 k^2) \pi} \Bigg) + \\
& \frac{1}{4} r m^d \left(\frac{8}{d \pi} + \frac{1}{(-3+d) r m^3} - \frac{32 k^2 r m^2}{(2+d) \pi (1+4 k^2 r m^2)} - \frac{16 d k^2 r m^2}{(2+d) \pi (1+4 k^2 r m^2)} - \right. \\
& \frac{\operatorname{Hypergeometric2F1}\left[-\frac{1}{2}, \frac{1}{2}(-3+d), \frac{1}{2}(-1+d), -k^2 n^2 r m^2\right]}{(-3+d) r m^3} + \\
& \frac{k^2 n^2 \operatorname{Hypergeometric2F1}\left[-\frac{1}{2}, \frac{1}{2}(-1+d), \frac{1+d}{2}, -k^2 n^2 r m^2\right]}{(-1+d) r m} - \\
& \frac{k^4 n^4 r m \operatorname{Hypergeometric2F1}\left[\frac{1}{2}, \frac{1+d}{2}, \frac{3+d}{2}, -k^2 n^2 r m^2\right]}{1+d} - \\
& \frac{32 k^2 r m^2 \operatorname{Hypergeometric2F1}\left[1, 1+\frac{d}{2}, 2+\frac{d}{2}, -4 k^2 r m^2\right]}{(2+d) \pi} + \\
& \frac{16 d k^2 r m^2 \operatorname{Hypergeometric2F1}\left[1, 1+\frac{d}{2}, 2+\frac{d}{2}, -4 k^2 r m^2\right]}{(2+d) \pi (1+4 k^2 r m^2)} + \\
& \left. \frac{64 d k^4 r m^4 \operatorname{Hypergeometric2F1}\left[1, 1+\frac{d}{2}, 2+\frac{d}{2}, -4 k^2 r m^2\right]}{(2+d) \pi (1+4 k^2 r m^2)} \right), \\
& (r m \notin \text{Reals} \mid \mid \operatorname{Re}[r m] \geq 0) \&\& \left(\operatorname{Im}\left[\frac{1}{k - k r m}\right] \geq 2 + 2 \operatorname{Re}\left[\frac{1}{-1 + r m}\right] \mid \mid \right. \\
& \left. \operatorname{Im}\left[\frac{1}{k - k r m}\right] \leq 2 \operatorname{Re}\left[\frac{1}{-1 + r m}\right] \mid \mid 2 \operatorname{Im}\left[\frac{1}{-1 + r m}\right] + \operatorname{Re}\left[\frac{1}{k - k r m}\right] \neq 0 \right) \&\& \\
& \left(\operatorname{Im}\left[\frac{1}{k n (-1 + r m)}\right] + \operatorname{Re}\left[\frac{1}{-1 + r m}\right] \geq 0 \mid \mid 1 + \operatorname{Im}\left[\frac{1}{k n (-1 + r m)}\right] + \operatorname{Re}\left[\frac{1}{-1 + r m}\right] \leq 0 \mid \mid \right. \\
& \left. \operatorname{Im}[r m] \neq \frac{1}{\operatorname{Re}\left[\frac{1}{k n (-1 + r m)}\right]} \right) \&\& \\
& \left(\operatorname{Im}\left[\frac{1}{k - k r m}\right] + 2 \operatorname{Re}\left[\frac{1}{-1 + r m}\right] \geq 0 \mid \mid 2 + \operatorname{Im}\left[\frac{1}{k - k r m}\right] + 2 \operatorname{Re}\left[\frac{1}{-1 + r m}\right] \leq 0 \mid \mid \right.
\end{aligned}$$

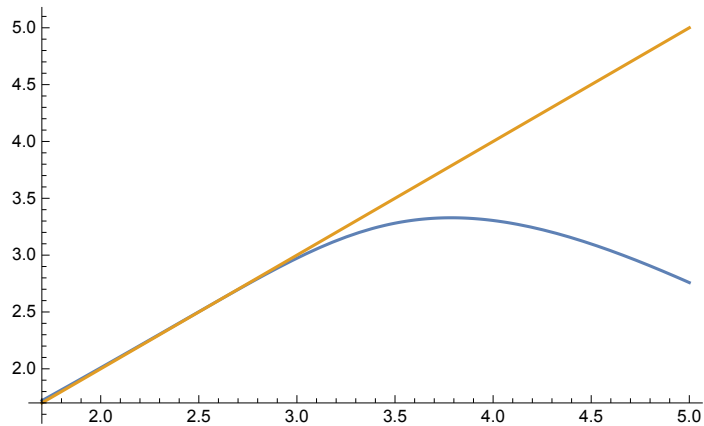
$$\begin{aligned}
& 2 \operatorname{Im}\left[\frac{1}{-1+rm}\right] \neq \operatorname{Re}\left[\frac{1}{k-krm}\right] \& \\
& \left(\operatorname{Im}\left[\frac{1}{-1+rm}\right] + \operatorname{Re}\left[\frac{1}{kn(-1+rm)}\right] \neq 0 \mid \mid \operatorname{Im}\left[\frac{1}{kn(-1+rm)}\right] \geq 1 + \operatorname{Re}\left[\frac{1}{-1+rm}\right] \mid \mid \right. \\
& \left. \operatorname{Im}\left[\frac{1}{kn(-1+rm)}\right] \leq \operatorname{Re}\left[\frac{1}{-1+rm}\right] \right)
\end{aligned}$$

I am getting pretty consistent results on sigma_n in live cell based on quite different methods. 1. From Yue Li's TEM data : For a dry cell sigma_n_dry = 0.04. SD~RI difference between two components. For a dry cell, this is between air and solid protein / DNA, which is $1.33 + 0.17 \text{ ml/g} \times 1.25 \text{ g/ml} - 1 = 0.54$. For a live cell, this is between water and the max RI = $0.17 \times 1.25 = 0.21$. Thus, sigma_n_live = sigma_n_dry * 0.21 / 0.54 = 0.0157. 2. From Wenli's modeling data : Density of nucleosomes divided by the density of pure protein = $1.66 \times 10^{(-27)} \times 1000 \times 300$, $000 / ((2 \times \pi \times (11 \times 10^{(-7)} / 2)^2 \times 6 \times 10^{(-7)})) / 1.25 = 35 \%$, where the molecular mass of nucleosome is 300 kD and I assumed it to be a cylinder with a height of 6 nm and a diameter of 11 nm. This is the volume fraction (v.f.) occupied by a single nucleosome. For a DNA bp, the volume fraction is 40 % (assuming that bp has a 1 nm radius). From Wenli's simulations, the average crowding in chromatin is 40 % but this assumes that the elementary elements are solid. If their internal volume fraction is only 35 - 40 % then the real volume fraction inside chromatin is $40 \% \times 35 \% = 14 \%$. This v.f. translates into an average RI of $1.33 + 0.17 \times 1.25 \times 0.14 = 1.36$, which is pretty consistent with what other groups have measured for cell nuclei. To estimate SD, assume that the PDF of density inside the nucleus is a uniform function between the minimal v.f. and the maximal v.f. The minimal v.f. should not be lower than the v.f. that would explain the RI of the media, which is 3.5 % (this is because RI of the media is 1.337). Thus, SD of intranuclear v.f. is = $7.38 \% = > \text{sigma_n} = 0.17 \times 1.25 \times 7.38 \% =$. This is the same answer as in method 1. 3. From prior quantitative phase contrast measurements made by other groups : Average RI of the nucleus for HeLa and a few other cell types is ~1.36. The v.f. that matches this RI is 14 % : $1.33 + 0.17 \times 1.25 \times 0.14 = 1.36$. If we again assume a uniform distribution of RI between the min and the max, the derivations made in method 2 apply and we again get sigma_n = . When I plugged in this sigma_n into the equation for Sigma that is based on a fractal ACF (not WM), I do get Sigma values pretty close to the experimentally observed ~0.03 - 0.07. The figure below is Sigma analytical as a function of D. The line is a linear fit around D = 2.33.

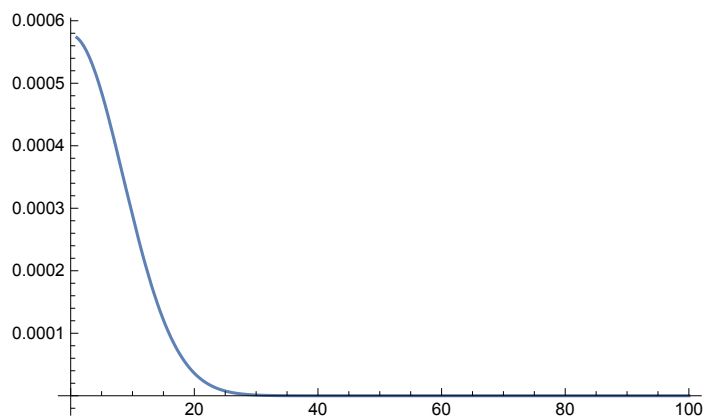
```

mf = 10000000;
p[r_, x_, a_, a1_, b_] := (r/x^(1/a))^b * Exp[-(r/x^(1/a))^a1] / x^(3/a);
mass[r_, a_, a1_, b_] := NIntegrate[
  (r1/x^(1/a))^b * Exp[-(r1/x^(1/a))^a1] / x^(3/a) * r1^2, {r1, 0, r}, {x, 1, mf}];
LogLogPlot[mass[r, 2.5, 1, 0], {r, 20, 100}];
dimension[a_] := Log[mass[51, a, 1, 2] / mass[50, a, 1, 2]] / Log[51/50];
Plot[{dimension[a], a}, {a, 1.7, 5}]

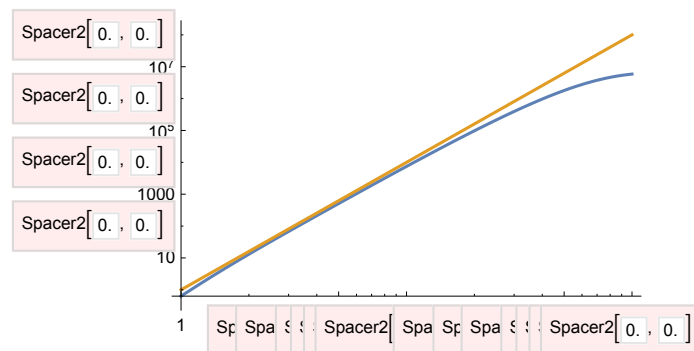
```



```
Plot[p[r, 500, 2.5, 2, 0], {r, 1, 100}, PlotRange -> All]
```



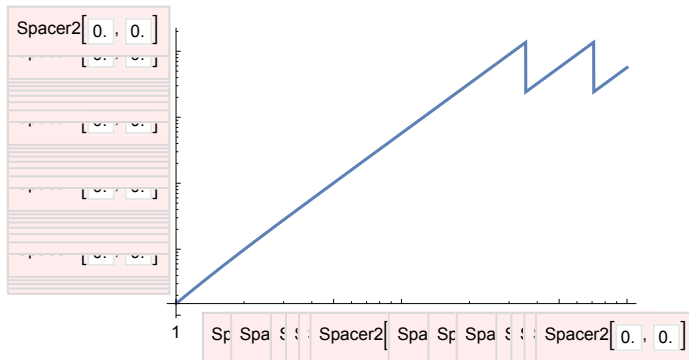
```
LogLogPlot[{mass[r, 4, 2, 2], r^4}, {r, 1, 100}]
```



```

mass3[r_, a_, a1_, b_] := NIntegrate[(r1/x^(1/a))^b *
  Exp[-(r1/x^(1/a))^a1]/x^(3/a) * r1^2, {r1, 0, r}, {x, 1, Infinity}];
LogLogPlot[{mass3[r, 2.5, 2, 0]}, {r, 1, 100}, PlotRange -> All]

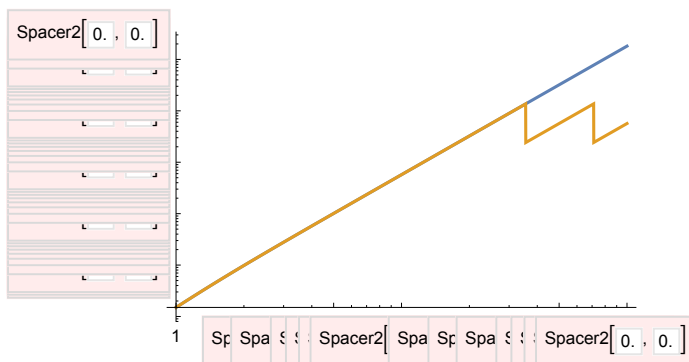
```



```

dens[r_, a_, a1_, b_] :=
  NIntegrate[(r/x^(1/a))^b * Exp[-(r/x^(1/a))^a1]/x^(3/a), {x, 1, Infinity}];
mass2[r_, a_, a1_, b_] := NIntegrate[dens[r1, a, a1, b] * r1^2, {r1, 0, r}];
LogLogPlot[{mass2[r, 2.5, 2, 0], mass3[r, 2.5, 2, 0]}, {r, 1, 100}, PlotRange -> All]

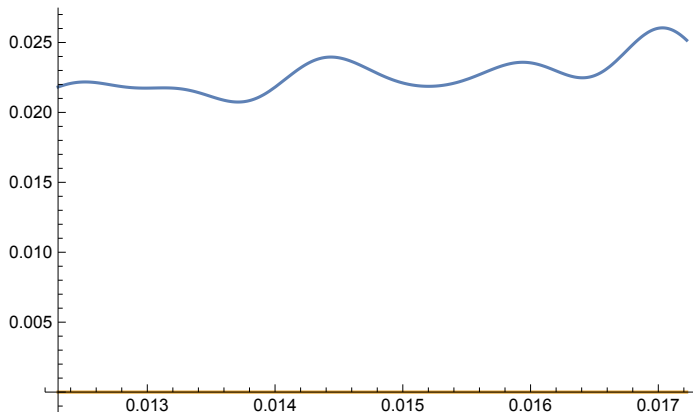
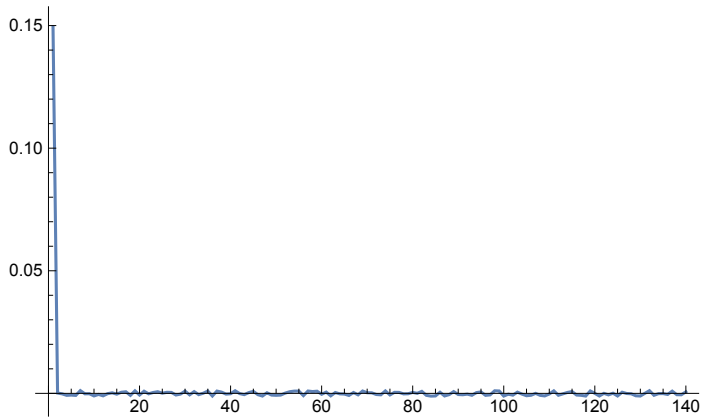
```



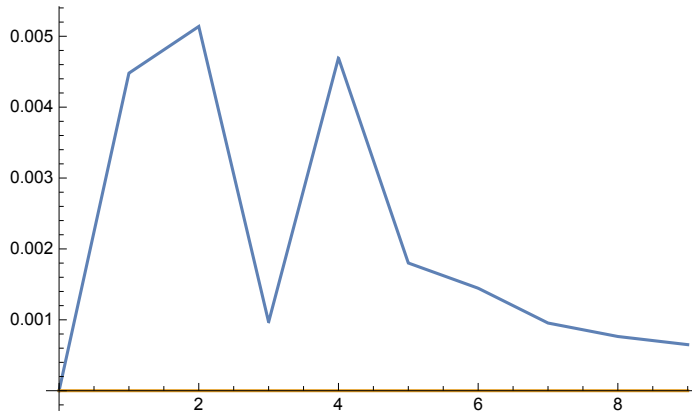

```

lc = 50;
na = 1;
rimax = 0.18 * 1.25 * 0.45 * 0.4 * (lc / (600 / 2 / na))^2;
rimean = 0.18 * 1.25 * 0.45 * 0.4;
reref = 1.52 - 1.33 - rimean;
thick = 7000;
num = 100;
k1 = 2 * Pi * (1.33 + rimean) / 700;
k2 = 2 * Pi * (1.33 + rimean) / 500;
ntab = Table[If[n == 1, reref, RandomReal[{-rimax, rimax}]], {n, 1, thick/lc}];
opd = Table[(n - 1) * lc, {n, 1, Length[ntab]}];
spec[k_] := Abs[Sum[ntab[[n]] * Exp[-I * k * opd[[n]]], {n, 1, Length[ntab]}]]^2;
spectab = Table[spec[k1 + (k2 - k1) * (n - 1) / (num - 1)], {n, 1, num}];
specosc = Table[spectab[[n]] - Mean[spectab], {n, Length[spectab]}];
psd = Abs[Fourier[specosc]];
psdi = Interpolation[psd, InterpolationOrder -> 1];
ListPlot[ntab, PlotJoined -> True, PlotRange -> All]
Plot[{spec[k], 0}, {k, k1, k2}]
numner = N[(k2 - k1) * thick / (2 * Pi)] - Mod[N[(k2 - k1) * thick / (2 * Pi)], 1] + 1
Plot[{psdi[x + 1], 0}, {x, 0, numner + 3}, PlotRange -> All]

```



6.



Attempt at analysing Kai ' s data :

```
rg = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/rg.txt",
  Number, RecordLists → True];
rgpdf = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/rgpdf.txt",
  Number, RecordLists → True];
phikai = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/phi.txt",
  Number, RecordLists → True];
alpha = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/alpha.txt",
  Number, RecordLists → True];

rg = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/rg.txt",
  Number, RecordLists → True];
rgpdf = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/rgpdfunnorm.txt",
  Number, RecordLists → True];
phikai = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/phi.txt",
  Number, RecordLists → True];
alpha = ReadList["C:/Users/vba990/Desktop/Math_data/Kai/alpha.txt",
  Number, RecordLists → True];
```

```

surfaceterm = 0;
massfixed = 1;
nachoise = 0;

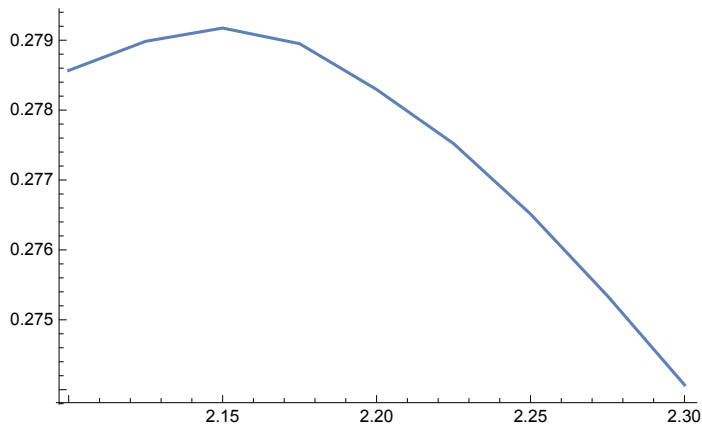
dexpand = 2.5;
mf = 1000000;
cvcdefault = 0.35;
phinucleoplasm = 0.1;

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = nailive;

rig = 1.517 + 1.53 * 0;
rim = 1.337;

riinc = 0.1899 * 0 + (0.1899 * 1 / 2 + 0.17 * 1 / 2 + 0.185 * 0);
purechromdensity = 0.555;
ri[x_] := 1.332 + riinc * purechromdensity * x;
sn[phi_] := Sqrt[phi * (1 - phi)] * (ri[1] - ri[phinucleoplasm]);
thick = lambda / 1000 * rig / 2 / nai^2;
ric[phi_] := ri[phi + (1 - phi) * phinucleoplasm];
size[lc_, d_, lmin_, lmax_] :=
  (3 - d) / (1 - (lmin / lmax)^(3 - d)) * (lmin / lc)^(4 - d) / lmin;
rcoeff[phi_] := (2 * rig / (rig + ric[phi]) * 2 * ric[phi] / (rig + ric[phi]) *
  (rig - ric[phi]) / (rig + ric[phi]) * (rig + rim)^2 / (rig - rim)^2)^2;
kt[l_, phi_] := 2 * Pi / lambda * ric[phi] * l * 1000;
xp[lc_, phi_] := 2 * Pi / lambda * ric[phi] * lc;
na = nachoise * Min[(1.4 + nai) / rig, 1] + (1 - nachoise) * 1.4 / rig;
s2[lc_, y_, phi_] :=
  2 / Pi * (kt[y, phi] * xp[lc, phi]^3 * na^2) / (1 + xp[lc, phi]^2 * (4 + na^2)) /
  (1 + xp[lc, phi]^2 * 4 + surfaceterm * 1 / 4 * (1 - 1 / Sqrt[1 + (xp[lc, phi] * na)^2]));
sigma[aindex_] := Sqrt[Sum[rgpdf[[i, aindex]] / 1000 * s2[rg[[i, 1]], thick, cvcdefault] *
  phikai[[i, aindex]] * (1 - phikai[[i, aindex]]), {i, 1, Length[rg]}] /
  Sum[rgpdf[[i, aindex]] / 1000 * phikai[[i, aindex]] * (1 - phikai[[i, aindex]]),
  {i, 1, Length[rg]}]];
sigmaexperiment = Table[If[j == 1, alpha[[i, 1]], sigma[i] * Sqrt[rcoeff[cvcdefault]] *
  sn[cvcdefault]], {i, 1, Length[alpha]}, {j, 1, 2}];
Show[ListPlot[sigmaexperiment, PlotJoined -> True]]

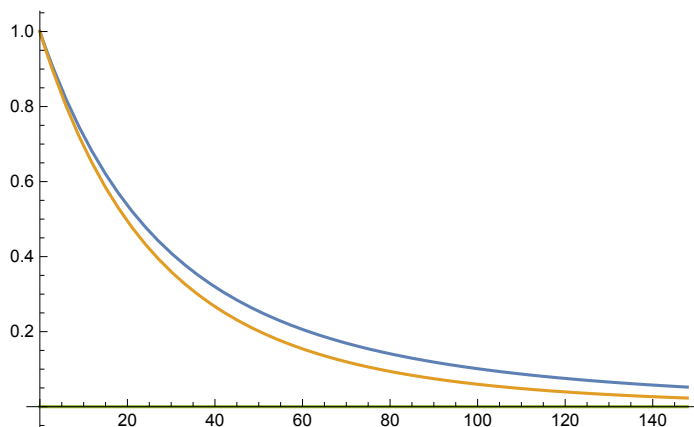
```



```
Text["Kai ACF is a stretch exp; alpha=2.1 => power =0.66; alpha=2.3 => power=0.7"]
acf[aindex_, r_] :=
```

```
Sum[rgpdf[[i, aindex]] / 1000 * Exp[-r / rg[[i, 1]]] * phikai[[i, aindex]] *
  (1 - phikai[[i, aindex]]), {i, 1, Length[rg]}] / Sum[rgpdf[[i, aindex]] / 1000 *
  phikai[[i, aindex]] * (1 - phikai[[i, aindex]]), {i, 1, Length[rg]}];
Plot[{acf[1, r], acf[Length[alpha], r], 0}, {r, 0, 0.3 * rg[[Length[rg], 1]]},
  PlotRange -> All]
3 + (Log[acf[1, 100]] - Log[acf[1, 20]]) / (Log[100] - Log[20])
3 + (Log[acf[2, 100]] - Log[acf[2, 20]]) / (Log[100] - Log[20])
3 + (Log[acf[9, 100]] - Log[acf[9, 20]]) / (Log[100] - Log[20])
LogLogPlot[{acf[1, r], acf[2, r], acf[Length[alpha], r], 0},
  {r, 0, 0.3 * rg[[Length[rg], 1]]}, PlotRange -> All]
Show[LogLogPlot[{-Log[acf[1, r]], -Log[acf[Length[alpha], r]], 0},
  {r, 0, 1 * rg[[Length[rg], 1]]}, PlotRange -> All],
  LogLogPlot[{(r / 25)^0.66, (r / 20)^0.7}, {r, 10, 300}]]
```

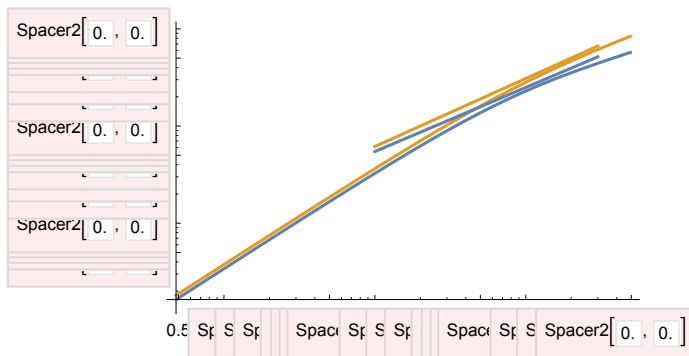
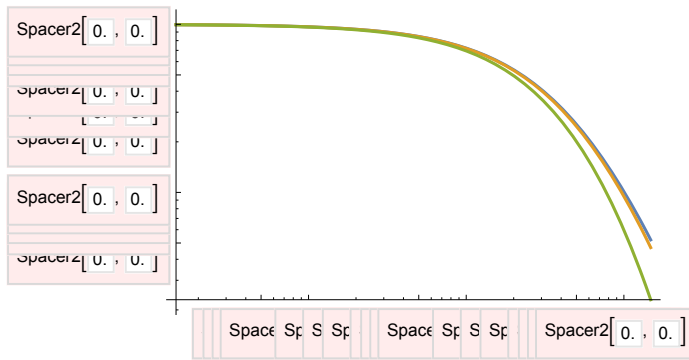
Kai ACF is a stretch exp; alpha=2.1 => power =0.66; alpha=2.3 => power=0.7



1.96348

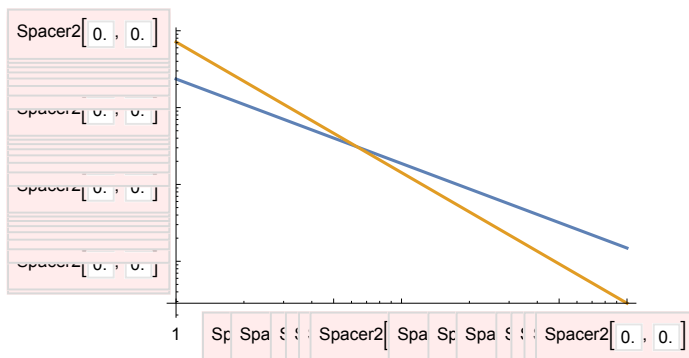
1.92843

1.68691



```
cvc[aindex_] := Sum[rgpdf[[i, aindex]] / 1000 *
  4 / 3 * Pi * rg[[i, 1]]^3 * phikai[[i, aindex]], {i, 1, Length[rg]}] /
  Sum[rgpdf[[i, aindex]] / 1000 * 4 / 3 * Pi * rg[[i, 1]]^3, {i, 1, Length[rg]}];
cvc[
  7]
0.0416708
```

```
size[lc_, d_, lmin_, lmax_] := (3 - d) / (1 - (lmin / lmax)^(3 - d)) * (lmin / lc)^(4 - d) / lmin;
LogLogPlot[{size[lc, 2.9, 1, 250], size[lc, 2.3, 1, 250]}, {lc, 1, 100}, PlotRange -> All]
```



```
Integrate[size[lc, d, lmin, lmax] * (lc/lmin)^3, {lc, lmin, lmax}]
```

```
ConditionalExpression[ $\frac{(3-d) \left(-1 + \left(\frac{lmin}{lmax}\right)^{-d}\right)}{d \left(1 - \left(\frac{lmin}{lmax}\right)^{3-d}\right)}$ ,  $\left(\frac{lmin}{lmax - lmin} \neq 0 \&\& \operatorname{Re}\left[\frac{lmin}{lmax - lmin}\right] \geq 0\right) \mid \mid$   

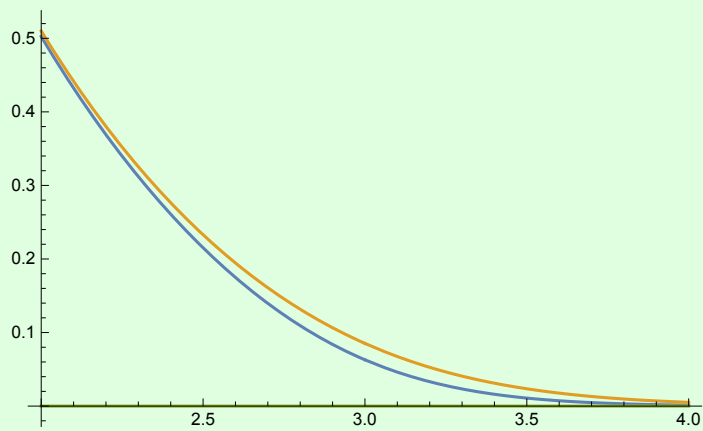
 $\operatorname{Re}\left[\frac{lmin}{lmax - lmin}\right] < -1 \mid \mid \frac{lmin}{lmax - lmin} \notin \operatorname{Reals}[Null]$ ]
```

```
x1 = 200;
```

```
x2 = 50;
```

```
Plot[{ $\frac{(3-d) * (1 - 1/x1^d)}{d \left(1 - \left(\frac{1}{x1}\right)^{3-d}\right)}$ ,  $\frac{(3-d) * (1 - 1/x2^d)}{d \left(1 - \left(\frac{1}{x2}\right)^{3-d}\right)}$ , 0}, {d, 2, 4}]
```

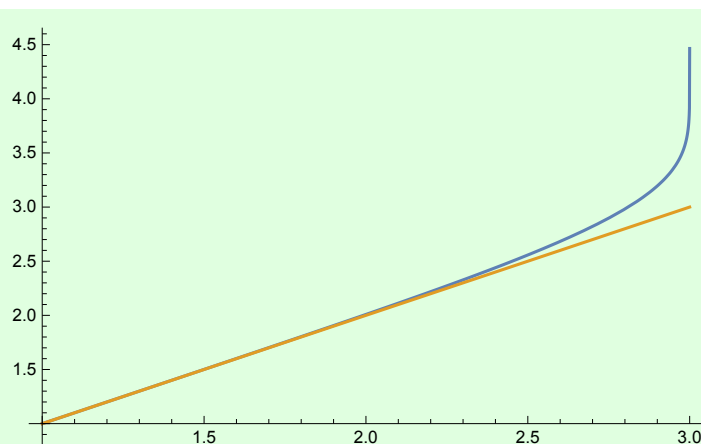
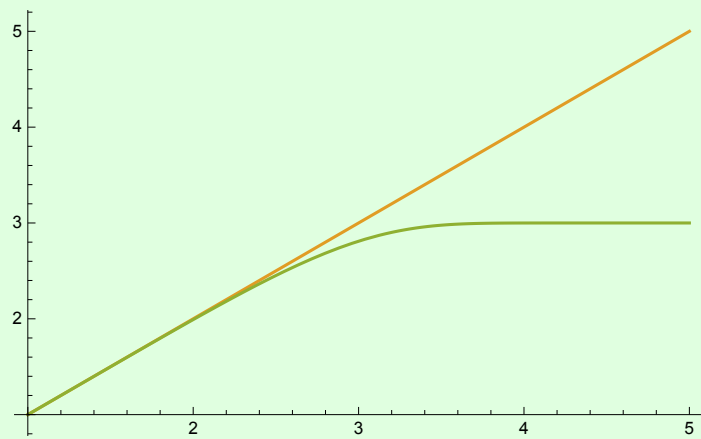
```
correction[d_] :=  $\frac{(3-d) * (1 - 1/100^d)}{d \left(1 - \left(\frac{1}{100}\right)^{3-d}\right)}$ ;
```



```

acf[x_, d_, lmin_, lmax_] :=
  ((3 - d) (
    (lmin^4 (lmin/lmax)^-d ExpIntegralE[-2 + d, x/lmax] - lmin ExpIntegralE[-2 + d, x/lmin]) /
    lmax^3
  )) /
  (lmin (1 - (lmin/lmax)^(3-d)));
delta = 0.1;
acfd[d_, lmin_, lmax_] := 3 + (Log[acf[(lmax + lmin)/100 + delta, d, lmin, lmax]] -
  Log[acf[(lmax + lmin)/100, d, lmin, lmax]]) /
  (Log[(lmax + lmin)/100 + delta] - Log[(lmax + lmin)/100]);
Plot[{acfd[d, 1, mass[d]^(1/d)], d, 3 * (1 - Exp[-(d/3)^7])^(1/7)}, {d, 1, 5}]
gamma = 7;
dsize[d_] := 3 * (Log[1 / (1 - (d/3)^gamma)])^(1/gamma);
trueacfd[d_] := acfd[d, 1, mass[d]^(1/d)];
Plot[{dsize[d], d}, {d, 1, 3}]

```



```

surfaceterm = 0;
massfixed = 1;
nachoise = 0;
masscorrection = 1;

```

```

dexpand = 2.5;
mf = 1000000;
cvcdefault = 0.35;
phinucleoplasma = 0.1;

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = nailive;

rig = 1.517 + 1.53 * 0;
rim = 1.337;

riinc = 0.1899 * 0 + (0.1899 * 1 / 2 + 0.17 * 1 / 2 + 0.185 * 0);
purechromdensity = 0.555;
ri[x_] := 1.332 + riinc * purechromdensity * x;
sn[phi_] := Sqrt[phi * (1 - phi)] * (ri[1] - ri[phinucleoplasma]);

ric[phi_] := ri[phi + (1 - phi) * phinucleoplasma];
size[lc_, d_, lmin_, lmax_] :=
  (3 - d) / (1 - (lmin / lmax)^(3 - d)) * (lmin / lc)^(4 - d) / lmin;
rcoeff[phi_] := (2 * rig / (rig + ric[phi]) * 2 * ric[phi] / (rig + ric[phi]) *
  (rig - ric[phi]) / (rig + ric[phi]) * (rig + rim)^2 / (rig - rim)^2)^2;
kt[l_, phi_] := 2 * Pi / lambda * ric[phi] * l * 1000;
xp[lc_, phi_] := 2 * Pi / lambda * ric[phi] * lc;
na = nachoice * Min[(1.4 + nai) / rig, 1] + (1 - nachoice) * 1.4 / rig;
s2[lc_, y_, phi_] :=
  2 / Pi * (kt[y, phi] * xp[lc, phi]^3 * na^2) / ((1 + xp[lc, phi]^2 * (4 + na^2)) /
  (1 + xp[lc, phi]^2 * 4) + surfaceterm * 1 / 4 * (1 - 1 / Sqrt[1 + (xp[lc, phi] * na)^2]));
sigma[d_, lmin_, lmax_, y_, phi_] := Sqrt[NIntegrate[(3 - d) / (1 - (lmin / lmax)^(3 - d)) *
  (lmin / lc)^(4 - d) / lmin * s2[lc, y, phi], {lc, lmin, lmax}]];
sigmanorm[d_, lmin_, lmax_, t_, phi_] := sigma[d, lmin, lmax, t, phi] *
  2 * Pi / (na * Sqrt[kt[t, phi]])^1;
sigmaexperiment[d_, lmin_, lmax_, t_, phi_] :=
  sigma[d, lmin, lmax, t, phi] * Sqrt[rcoeff[phi]] * sn[phi];
sigmathinfilmm[d_] := Sqrt[2] * rcoeff[phi]^0.5 * (ric[phi] - rim) / (ric[phi] + rim);
sigmaexperimentwithinfilm[d_, lmin_, lmax_, t_, phi_] :=
  Sqrt[sigmaexperiment[d, lmin, lmax, t, phi]^2 + sigmathinfilmm[d]^2];

thick = lambda / 1000 * rig / 2 / nai^2;

lmaxlminapprox = 100;
correction[d_] := 
$$\frac{(3 - d) * (1 - 1 / \text{lmaxlminapprox}^d)}{d \left(1 - \left(\frac{1}{\text{lmaxlminapprox}}\right)^{3-d}\right)}$$
;
mass[d_] := (mf * massfixed + (1 - massfixed) * 4 * 10^(-7) * Exp[11.1 * d]) /
  correction[d]^masscorrection;

```



```

"Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default"
Text["D_expand=", dexpand, "CVC default=", cvcdefault]
a4 = sigmaexperiment[dexpand, 1, mass[dexpand]^(1/dexpand), thick, cvcdefault];
b4 = 1/D[sigmaexperiment[d, 1, mass[d]^(1/d), thick, cvcdefault], d] /. d -> dexpand
c4 = N[(-a4 + 1/b4 * dexpand) * b4]
df[d_] := d + 0 * If[d <= 3, d, 3];
Show[Plot[{sigmaexperiment[d, 1, mass[df[d]]^(1/df[d]), thick, cvcdefault], 0},
  {d, 1, 5}, PlotRange -> All], Plot[{1/b4 * (d - c4)}, {d, 2, 3.5}, PlotRange -> All]]

gamma = 7;
dsize[d_] := 3 * (Log[1/(1 - (d/3)^gamma)])^(1/gamma);
trueacfd[d_] := acfd[d, 1, mass[d]^(1/d)];
a5 = sigmaexperiment[dsize[dexpand], 1,
  mass[dsize[dexpand]]^(1/dsize[dexpand]), thick, cvcdefault];
b5 = 1/D[sigmaexperiment[dsize[d], 1, mass[dsize[d]]^(1/dsize[d]),
  thick, cvcdefault], d] /. d -> dexpand
c5 = N[(-a5 + 1/b5 * dexpand) * b5]
Show[Plot[
  {sigmaexperiment[dsize[d], 1, mass[dsize[d]]^(1/dsize[d]), thick, cvcdefault], 0},
  {d, 1, 3}, PlotRange -> All],
  Plot[{1/b5 * (d - c5), 1/b4 * (dsize[d] - c4)}, {d, 2, 3}, PlotRange -> All]]

```

Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default

Text[D_expand=, 2.5, CVC default=, 0.35]

NIntegrate: $lc = 1. \times 10^6 \frac{1}{d} \left(\frac{(1. - 1. \times 100.^{-3.+d}) d}{(1. - 1. \times 100.^{\text{Times}[\ll 2 \gg]} (3. - 1. d))} \right)^{\frac{1}{d}}$ is not a valid limit of integration.

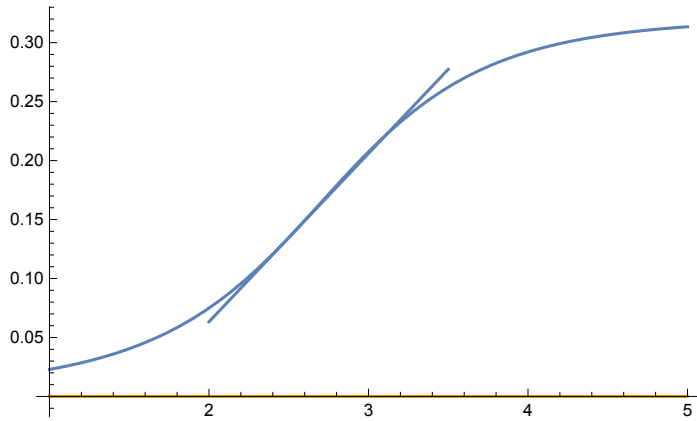
NIntegrate: $lc = 1. \times 10^6 \frac{1}{d} \left(\frac{(1. - 1. \times 100.^{-3.+d}) d}{(1. - 1. \times 100.^{\text{Times}[\ll 2 \gg]} (3. - 1. d))} \right)^{\frac{1}{d}}$ is not a valid limit of integration.

NIntegrate: $lc = 1. \times 10^6 \frac{1}{d} \left(\frac{(1. - 1. \times 100.^{-3.+d}) d}{(1. - 1. \times 100.^{\text{Times}[\ll 2 \gg]} (3. - 1. d))} \right)^{\frac{1}{d}}$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

7.00624

1.55656



NIntegrate: $lc = 3, \frac{\frac{0.333333}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}}{\log\left[\frac{1}{1-\ll 22 \gg \ll 1 \gg}\right]^{1/7}} \times 100, \frac{\left(\left(1 - 1. \times 100.^{-3+\text{Times}[\ll 2 \gg]}\right) \log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}\right)}{\left(1 - 1. \times 100.^{\text{Times}[\ll 2 \gg]}\right) (3 - 3. \log[\ll 1 \gg]^{1/7})} \frac{\frac{0.333333}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}$ is not a valid

limit of integration.

NIntegrate: $lc = 3, \frac{\frac{0.333333}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}}{\log\left[\frac{1}{1-\ll 22 \gg \ll 1 \gg}\right]^{1/7}} \times 100, \frac{\left(\left(1 - 1. \times 100.^{-3+\text{Times}[\ll 2 \gg]}\right) \log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}\right)}{\left(1 - 1. \times 100.^{\text{Times}[\ll 2 \gg]}\right) (3 - 3. \log[\ll 1 \gg]^{1/7})} \frac{\frac{0.333333}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}$ is not a valid

limit of integration.

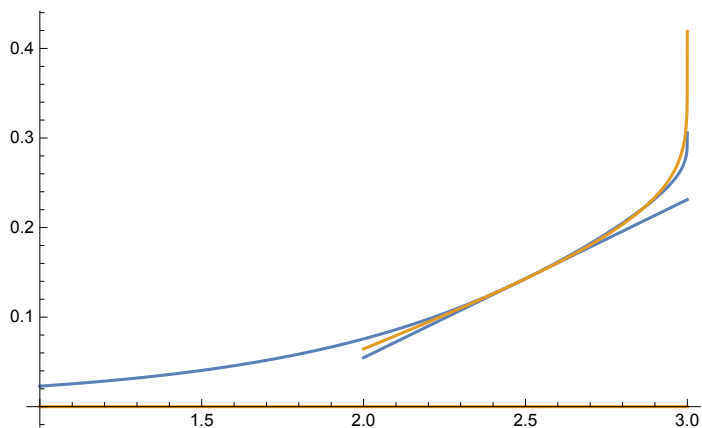
NIntegrate: $lc = 3, \frac{\frac{0.333333}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}}{\log\left[\frac{1}{1-\ll 22 \gg \ll 1 \gg}\right]^{1/7}} \times 100, \frac{\left(\left(1 - 1. \times 100.^{-3+\text{Times}[\ll 2 \gg]}\right) \log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}\right)}{\left(1 - 1. \times 100.^{\text{Times}[\ll 2 \gg]}\right) (3 - 3. \log[\ll 1 \gg]^{1/7})} \frac{\frac{0.333333}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}}{\log\left[\frac{1}{1+\text{Times}[\ll 2 \gg]}\right]^{1/7}}$ is not a valid

limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

5.66376

1.69032



```

InverseLaplaceTransform[1/s^(3-d)*Exp[-s/a], s, t]
InverseLaplaceTransform[1/s^(3-d), s, t]
InverseLaplaceTransform[1/(s+b)^(3-d)*Exp[-s/a], s, t]

$$\frac{\left(-\frac{1}{a}+t\right)^{2-d} \text{HeavisideTheta}\left[-\frac{1}{a}+t\right]}{\Gamma[3-d]}$$


$$\frac{t^{2-d}}{\Gamma[3-d]}$$


$$\frac{e^{-b\left(-\frac{1}{a}+t\right)}\left(-\frac{1}{a}+t\right)^{2-d} \text{HeavisideTheta}\left[-\frac{1}{a}+t\right]}{\Gamma[3-d]}$$

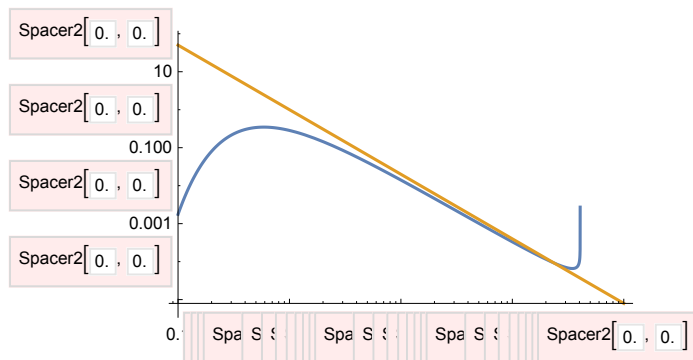
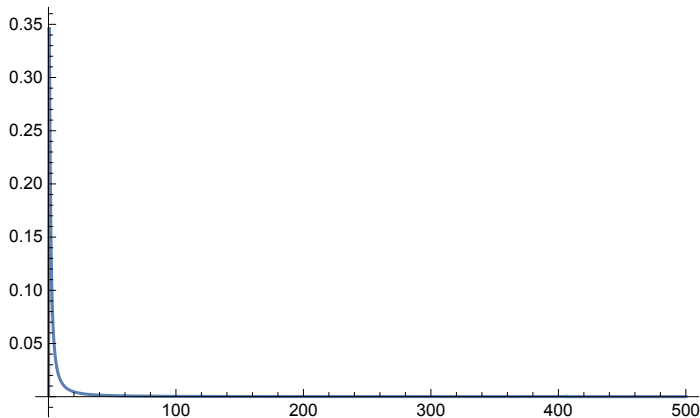

```

```

Text["Contribution function (lc) =
  LT(1/x)*1/x^2 = 1/lc^(4-D) up to lc=Rmax in the Exp[-r/Rmax]"
lt[t_, d_, a_] := 
$$\frac{\left(-\frac{1}{a} + t\right)^{2-d} \text{HeavisideTheta}\left[-\frac{1}{a} + t\right]}{\text{Gamma}[3 - d]}$$
;
lt0[t_, d_] := 
$$\frac{t^{2-d}}{\text{Gamma}[3 - d]}$$
;
ltmin[t_, d_, b_] := 
$$\frac{e^{-b t} t^{2-d}}{\text{Gamma}[3 - d]}$$
;
ltminmax[t_, d_, b_, a_] := 
$$\frac{e^{-b \left(-\frac{1}{a} + t\right)} \left(-\frac{1}{a} + t\right)^{2-d} \text{HeavisideTheta}\left[-\frac{1}{a} + t\right]}{\text{Gamma}[3 - d]}$$
;
dist[t_, d_, b_, a_] := ltminmax[1/t, d, b, a] * 1/t^2;
Plot[{dist[t, 2.3, 1, 1000000^(1/2.3)]}, {t, 0.1, 500}, PlotRange -> All]
LogLogPlot[{dist[t, 2.3, 1, 1000000^(1/2.3)]}, 1/t^(4 - 2.3)},
{t, 0.1, 1000}, PlotRange -> All]

```

Contribution function (lc) = LT(1/x)*1/x^2 = 1/lc^(4-D) up to lc=Rmax in the Exp[-r/Rmax]



$\text{InverseLaplaceTransform}\left[\frac{\text{Exp}[-r/rc]}{(1+r/rmin)^{(3-d)}}, r, t\right]$

$$\frac{e^{-rmin \left(-\frac{1}{rc} + t\right)} rmin \left(rmin \left(-\frac{1}{rc} + t\right)\right)^{2-d} \text{HeavisideTheta}\left[-\frac{1}{rc} + t\right]}{\text{Gamma}[3 - d]}$$

```
Integrate[Exp[-r1/rc] / (1 + r1/rmin)^(3 - d) * r1^2, {r1, 0, r}]
```

```
ConditionalExpression[ $\frac{1}{rc^2} e^{\frac{rmin}{rc}} rmin^3 \left(\frac{rmin}{rc}\right)^{-d}$   

 $\left(rmin^2 \Gamma\left[-2 + d, \frac{rmin}{rc}\right] + rc \left(-2 rmin \Gamma\left[-1 + d, \frac{rmin}{rc}\right] + rc \Gamma\left[d, \frac{rmin}{rc}\right]\right) -$   

 $\frac{1}{rc^2} e^{\frac{rmin}{rc}} rmin^3 \left(\frac{r + rmin}{rc}\right)^{-d} \left(\frac{r + rmin}{rmin}\right)^d$   

 $\left(rmin^2 \Gamma\left[-2 + d, \frac{r + rmin}{rc}\right] + rc \left(-2 rmin \Gamma\left[-1 + d, \frac{r + rmin}{rc}\right] + rc \Gamma\left[d, \frac{r + rmin}{rc}\right]\right) \right],$   

 $\left(\left(\text{Im}[rc] \text{Re}[r] \geq \text{Im}[r] \text{Re}[rc] \ \&\& \ (\text{Im}[rmin] \text{Re}[r] \geq \text{Im}[r] \text{Re}[rmin] \ || \ \text{Im}[rmin] \text{Re}[rc] \leq\right.\right.$   

 $\left.\left.\text{Im}[rc] \text{Re}[rmin] \ || \ \text{Im}[rc] (\text{Re}[r] + \text{Re}[rmin]) \leq (\text{Im}[r] + \text{Im}[rmin]) \text{Re}[rc]\right) \ || \right.$   

 $\left.(\text{Im}[rc] \text{Re}[r] \leq \text{Im}[r] \text{Re}[rc] \ \&\& \ (\text{Im}[rmin] \text{Re}[rc] \geq \text{Im}[rc] \text{Re}[rmin] \ || \ \text{Im}[rc] (\text{Re}[r] +\right.$   

 $\left.\text{Re}[rmin]) \geq (\text{Im}[r] + \text{Im}[rmin]) \text{Re}[rc] \ || \ \text{Im}[rmin] \text{Re}[r] \leq \text{Im}[r] \text{Re}[rmin])\right) \ \&\&$   

 $\left.\left(\left(\text{Re}\left[\frac{rmin}{r}\right] \geq 0 \ \&\& \ \frac{rmin}{r} \neq 0\right) \ || \ \frac{rmin}{r} \notin \text{Reals}[Null] \ || \ \text{Re}\left[\frac{rmin}{r}\right] < -1\right) \right]$ 
```

```
sizedist[t_, d_, rmin_, rc_] :=
```

```
 $\frac{1}{\Gamma[3 - d]} e^{-rmin \left(-\frac{1}{rc} + 1/t\right)} rmin \left(rmin \left(-\frac{1}{rc} + 1/t\right)\right)^{2-d} \text{HeavisideTheta}\left[-\frac{1}{rc} + 1/t\right] * 1/t^2;$ 
```

```
acf2[r_, d_, lmin_, lmax_] := Exp[-r/lmax] / (1 + r/lmin)^(3 - d);
```

```
massscaling[r_, d_, rmin_, rc_] :=  $\frac{1}{rc^2} e^{\frac{rmin}{rc}} rmin^3 \left(\frac{rmin}{rc}\right)^{-d}$   

 $\left(rmin^2 \Gamma\left[-2 + d, \frac{rmin}{rc}\right] + rc \left(-2 rmin \Gamma\left[-1 + d, \frac{rmin}{rc}\right] + rc \Gamma\left[d, \frac{rmin}{rc}\right]\right) -$   

 $\frac{1}{rc^2} e^{\frac{rmin}{rc}} rmin^3 \left(\frac{r + rmin}{rc}\right)^{-d} \left(\frac{r + rmin}{rmin}\right)^d \left(rmin^2 \Gamma\left[-2 + d, \frac{r + rmin}{rc}\right] +$   

 $rc \left(-2 rmin \Gamma\left[-1 + d, \frac{r + rmin}{rc}\right] + rc \Gamma\left[d, \frac{r + rmin}{rc}\right]\right);$ 
```

```
totalmass[d_, lmin_, lmax_] := NIntegrate[sizedist[t, d, lmin, lmax] * (t/lmin)^3,  

{t, lmin, lmax}];
```

```
LogLogPlot[{sizedist[t, 2.8, 1, 200], 1/t^(4 - 2.8)}, {t, 0, 220}, PlotRange -> All]
```

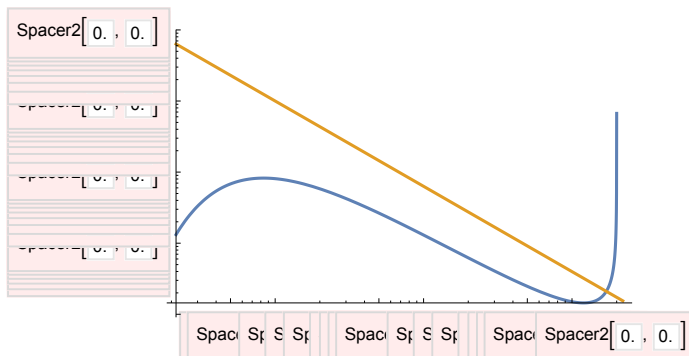
```
LogLogPlot[{acf2[r, 2.8, 1, 300], 1/r^(3 - 2.8)}, {r, 0.5, 500}]
```

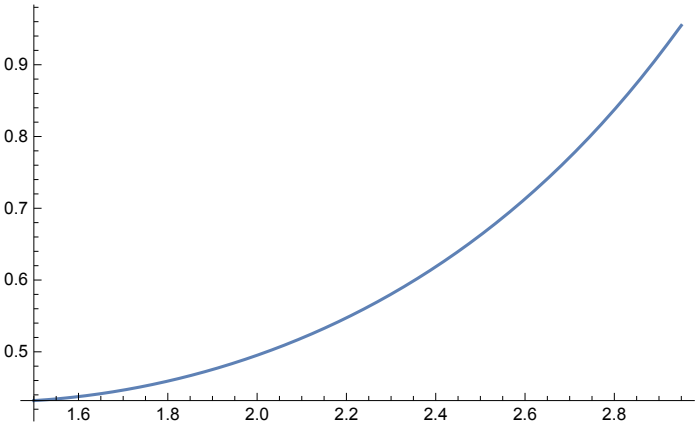
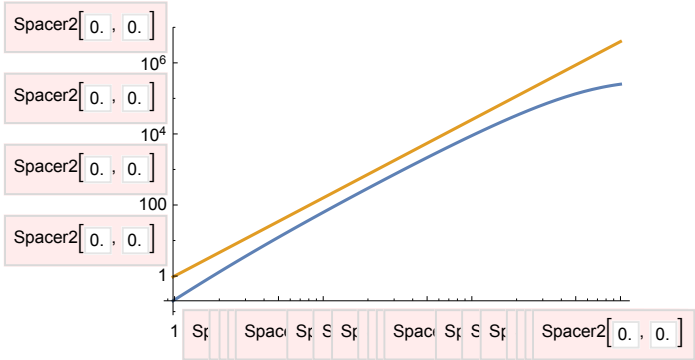
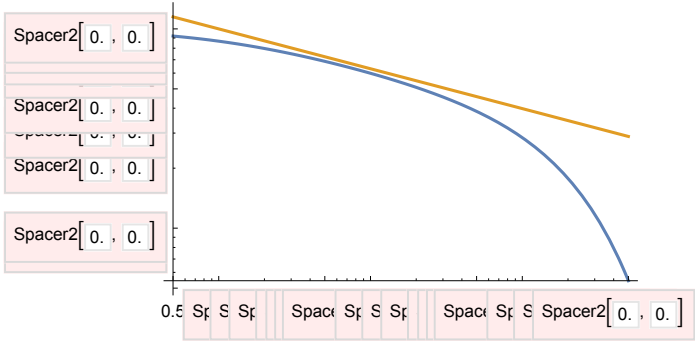
```
LogLogPlot[{massscaling[r, 2.2, 1, 300], r^2.2}, {r, 0, 1000}]
```

```
Plot[{totalmass[d, 1, 100] / 100^d}, {d, 1.5, 2.95}]
```

```
x1 = 100;
```

```
masscorrection[d_] := totalmass[d, 1, x1] / x1^d;
```

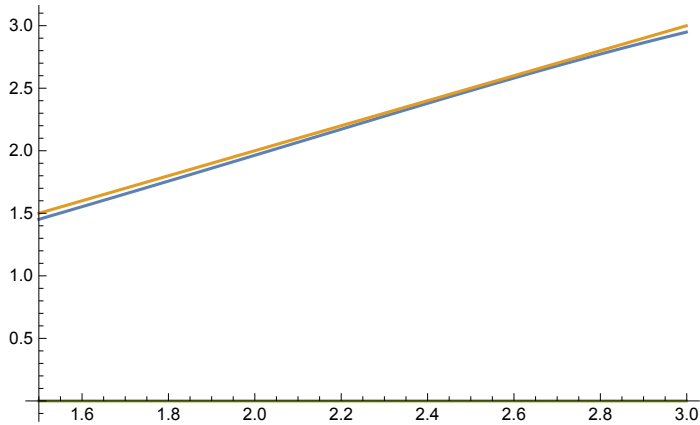




```

mf = 1000000;
delta = 0.1;
acfd2[d_, lmin_, lmax_] :=
  3 + (Log[acf2[(lmax + lmin) / 20 + delta, d, lmin, lmax]] - Log[acf2[(lmax + lmin) / 20, d,
    lmin, lmax]]) / (Log[(lmax + lmin) / 20 + delta] - Log[(lmax + lmin) / 20]);
Plot[{acfd2[d, 1, (mf / masscorrection[d])^(1/d)], d, 0}, {d, 1.5, 2.999}]

```



```

surfaceterm = 0;
massfixed = 1;
nachoise = 0;
masscorrectioncoef = 1;

```

```

dexpand = 2.5;
mf = 1000000;
cvcdefault = 0.35;
phinucleoplasm = 0.1;

```

```

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = nailive;

```

```

rig = 1.517 + 1.53 * 0;
rim = 1.337;

```

```

riinc = 0.1899 * 0 + (0.1899 * 1 / 2 + 0.17 * 1 / 2 + 0.185 * 0);
purechromdensity = 0.555;
ri[x_] := 1.332 + riinc * purechromdensity * x;
sn[phi_] := Sqrt[phi * (1 - phi)] * (ri[1] - ri[phinucleoplasm]);

```

```

ric[phi_] := ri[phi + (1 - phi) * phinucleoplasm];
size[lc_, d_, lmin_, lmax_] :=
  (3 - d) / (1 - (lmin / lmax)^(3 - d)) * (lmin / lc)^(4 - d) / lmin;
rcoeff[phi_] := (2 * rig / (rig + ric[phi]) * 2 * ric[phi] / (rig + ric[phi]) *
  (rig - ric[phi]) / (rig + ric[phi]) * (rig + rim)^2 / (rig - rim)^2)^2;
kt[l_, phi_] := 2 * Pi / lambda * ric[phi] * l * 1000;
xp[lc_, phi_] := 2 * Pi / lambda * ric[phi] * lc;

```

```

na = nachoice * Min[(1.4 + nai) / rig, 1] + (1 - nachoice) * 1.4 / rig;
s2[lc_, y_, phi_] :=
  2 / Pi * (kt[y, phi] * xp[lc, phi]^3 * na^2) / (1 + xp[lc, phi]^2 * (4 + na^2)) /
    (1 + xp[lc, phi]^2 * 4) + surfaceterm * 1 / 4 * (1 - 1 / Sqrt[1 + (xp[lc, phi] * na)^2]);
sizedist[t_, d_, rmin_, rc_] :=  $\frac{1}{\Gamma[3 - d]} e^{-rmin \left(-\frac{1}{rc} + 1/t\right)} rmin$ 
  (rmin (-1/rc + 1/t))^(2-d) HeavisideTheta[-1/rc + 1/t] * 1/t^2;
totalmass[d_, lmin_, lmax_] := NIntegrate[sizedist[t, d, lmin, lmax] * (t / lmin)^3,
  {t, 0, lmax}];
lmaxlminapprox = 100;
masscorrection[d_] := totalmass[d, 1, lmaxlminapprox] / lmaxlminapprox^d;
sigma[d_, lmin_, lmax_, y_, phi_] :=
  Sqrt[NIntegrate[sizedist[lc, d, lmin, lmax] * s2[lc, y, phi], {lc, lmin, lmax}]];
sigmanorm[d_, lmin_, lmax_, t_, phi_] :=
  sigma[d, lmin, lmax, t, phi] * 2 * Pi / (na * Sqrt[kt[t, phi]])^1;
sigmaexperiment[d_, lmin_, lmax_, t_, phi_] :=
  sigma[d, lmin, lmax, t, phi] * Sqrt[rcoeff[phi]] * sn[phi];
sigmathinfilm[phi_] := Sqrt[2] * rcoeff[phi]^0.5 * (ric[phi] - rim) / (ric[phi] + rim);
sigmaexperimentwithinfilm[d_, lmin_, lmax_, t_, phi_] :=
  Sqrt[sigmaexperiment[d, lmin, lmax, t, phi]^2 + sigmathinfilm[phi]^2];

thick = lambda / 1000 * rig / 2 / nai^2;

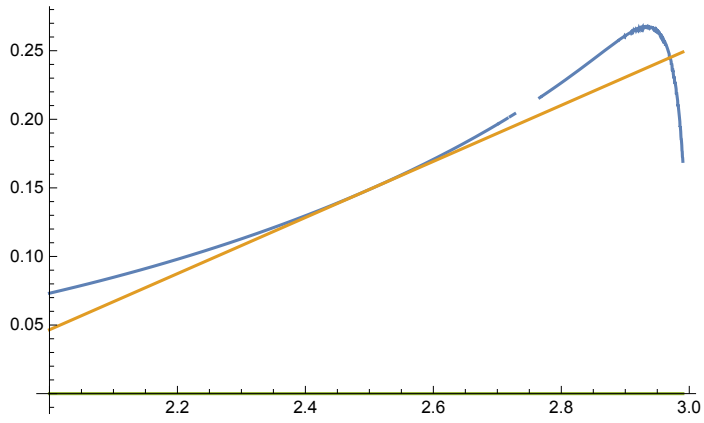
mass[d_] := (mf * massfixed + (1 - massfixed) * 4 * 10^(-7) * Exp[11.1 * d]) /
  masscorrection[d]^masscorrectioncoef;

"Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default"
Text["D_expand=", dexpand, "CVC default=", cvcdefault]
a = sigmaexperiment[dexpand, 1, mass[dexpand]^(1/dexpand), thick, cvcdefault];
deltad = 0.001;
b = deltad / (sigmaexperiment[dexpand + deltad, 1,
  mass[dexpand + deltad]^(1 / (dexpand + deltad)), thick, cvcdefault] -
  sigmaexperiment[dexpand, 1, mass[dexpand]^(1/dexpand), thick, cvcdefault])
c = N[(-a + 1 / b * dexpand) * b]
Plot[
  {sigmaexperiment[d, 1, mass[d]^(1/d), thick, cvcdefault], 1 / b * (d - c), 0}, {d, 2, 2.99}]
Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default
Text[D_expand=, 2.5, CVC default=, 0.35]

4.88873

1.77237

```

Sigma in WM case :

```

acf[r_, d_, rmin_, lc_] :=
  (r/rmin) ^ ((d - 3) / 2) * BesselK[(d - 3) / 2, r / lc] / BesselK[(d - 3) / 2, rmin / lc];
mass1[r_, d_, rmin_, lc_] := 
$$\frac{1}{3 d \text{BesselK}\left[\frac{1}{2}(-3 + d), \frac{rmin}{lc}\right]}$$


$$2^{-\frac{1}{2}-\frac{d}{2}} \pi r^2 \left(\frac{r}{rmin}\right)^{\frac{1}{2}(-1+d)} rmin \text{Abs}[lc]^{-\frac{3}{2}-\frac{d}{2}} \text{Abs}[r]^{\frac{1}{2}(-3-d)}$$


$$\left(2^d d \text{Abs}[lc]^d \text{Abs}[r]^3 \text{Gamma}\left[\frac{1}{2}(-3 + d)\right] \text{HypergeometricPFQ}\left[\left\{\frac{3}{2}\right\}, \left\{\frac{5}{2}, \frac{5}{2} - \frac{d}{2}\right\}, \frac{r^2}{4 lc^2}\right] + \right.$$


$$\left.24 \text{Abs}[lc]^3 \text{Abs}[r]^d \text{Gamma}\left[\frac{3}{2} - \frac{d}{2}\right] \text{HypergeometricPFQ}\left[\left\{\frac{d}{2}\right\}, \left\{-\frac{1}{2} + \frac{d}{2}, 1 + \frac{d}{2}\right\}, \frac{r^2}{4 lc^2}\right]\right);$$

mass2[r_, d_, rmin_, lc_] := 4 / 3 * rmin^3 * Pi * (r/rmin) ^ d *

$$2^{\wedge}((d - 5) / 2) / \text{BesselK}\left[\frac{1}{2}(-3 + d), \frac{rmin}{lc}\right] * (lc / rmin)^{\wedge}((3 - d) / 2) *$$

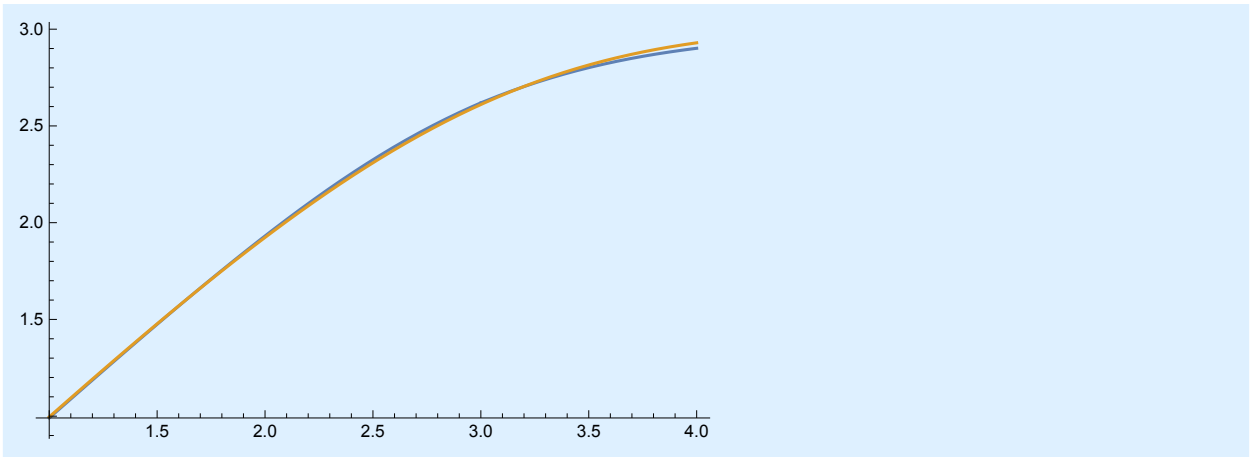
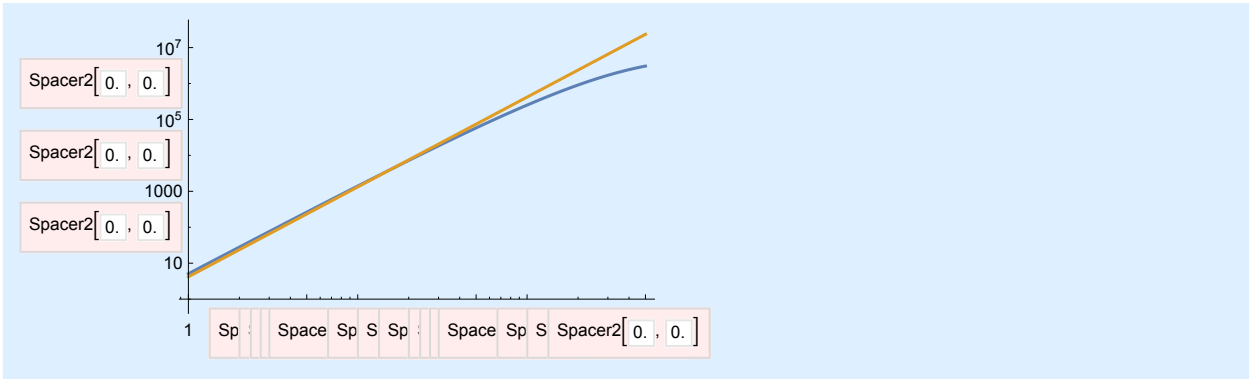

$$\left(\text{Gamma}\left[\frac{1}{2}(-3 + d)\right] \text{HypergeometricPFQ}\left[\left\{\frac{3}{2}\right\}, \left\{\frac{5}{2}, \frac{5}{2} - \frac{d}{2}\right\}, \frac{r^2}{4 lc^2}\right] * (r / lc)^{\wedge}(3 - d) + \right.$$


$$\left.2^{\wedge}(3 - d) * 3 / d * \text{Gamma}\left[\frac{3}{2} - \frac{d}{2}\right] \text{HypergeometricPFQ}\left[\left\{\frac{d}{2}\right\}, \left\{-\frac{1}{2} + \frac{d}{2}, 1 + \frac{d}{2}\right\}, \frac{r^2}{4 lc^2}\right]\right);$$

delta = 0.1;
w = 10;
fun[d_] := If[d <= 3, 3 - 2 * (d - 2), 1 + 5 * (d - 3)];
rmax[d_, nf_, rmin_] := rmin * (nf / fun[d]) ^ (1 / Min[d, 3]);
massd[d_, nf_, rmin_] :=
  (Log[mass2[(rmax[d, nf, rmin] + rmin * w) / w + delta, d, rmin, rmax[d, nf, rmin]]] -
    Log[mass2[(rmax[d, nf, rmin] + rmin * w) / w, d, rmin, rmax[d, nf, rmin]]]) /
    (Log[(rmax[d, nf, rmin] + rmin * w) / w + delta] - Log[(rmax[d, nf, rmin] + rmin * w) / w]);

d1 = 2.5;
mf = 200000;
g = 3.3;
LogLogPlot[{mass2[s, d1, 1, 200], 4 / 3 * Pi * s ^ Min[d1, 3]}, {s, 1, 500}]
Plot[{massd[d, mf, 1], 3 * (1 - Exp[-(d / 3) ^ g]) ^ (1 / g)}, {d, 1, 4}]

```

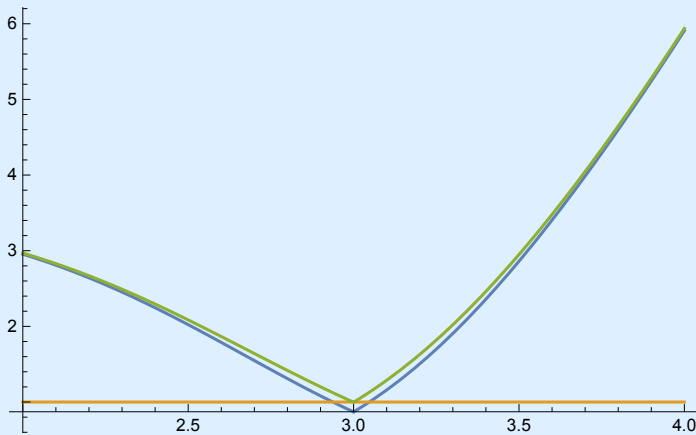


```

mass3[r_, d_, rmin_, lc_] :=
  (Gamma[1/2 (-3 + d)] HypergeometricPFQ[{3/2}, {5/2, 5/2 - d/2}, r^2/(4 lc^2)] * (r / lc)^(3 - d) +
    2^(3 - d) * 3 / d * Gamma[3/2 - d/2] HypergeometricPFQ[{d/2}, {-1/2 + d/2, 1 + d/2}, r^2/(4 lc^2)]);

d1 = 4;
LogLogPlot[{mass3[r, d1, 1, 200] * (r / 200)^d1, (r / 200)^3},
  {r, 1, 2000}, PlotRange -> All];
lm[d_, rmin_, lc_] := (lc / rmin)^d * (lc / rmin)^( (3 - d) / 2) *
  2^((d - 5) / 2) * (9.42 + (d - 3)^2) / BesselK[1/2 (-3 + d), rmin / lc];
lmd[d_, rmin_, lc_] := lm[d, rmin, lc] / (lc / rmin)^d;
Plot[{lmd[d, 1, 200] * 200^Max[d - 3, 0], 1, lmd[d, 1, 100] * 100^Max[d - 3, 0]}, {d, 2, 4}]

```



```

surfaceterm = 0;
massfixed = 1;
nachoise = 0;

dexpand = 2.5;
mf = 400000;
cvcdefault = 0.35;
phinucleoplasm = 0.1;

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = nailive;

rig = 1.517 + 1.53 * 0;
rim = 1.337;

riinc = 0.1899 * 0 + (0.1899 * 1 / 2 + 0.17 * 1 / 2 + 0.185 * 0);
purechromdensity = 0.555;

```

```

ri[x_] := 1.332 + riinc * purechromdensity * x;
sn[phi_] := Sqrt[phi * (1 - phi)] * (ri[1] - ri[phinucleoplasm]);

ric[phi_] := ri[phi + (1 - phi) * phinucleoplasm];
size[lc_, d_, lmin_, lmax_] :=
  (3 - d) / (1 - (lmin / lmax) ^ (3 - d)) * (lmin / lc) ^ (4 - d) / lmin;
rcoeff[phi_] := (2 * rig / (rig + ric[phi]) * 2 * ric[phi] / (rig + ric[phi]) *
  (rig - ric[phi]) / (rig + ric[phi]) * (rig + rim) ^ 2 / (rig - rim) ^ 2) ^ 2;
kt[l_, phi_] := 2 * Pi / lambda * ric[phi] * l * 1000;
xp[lc_, phi_] := 2 * Pi / lambda * ric[phi] * lc;
na = nachoice * Min[(1.4 + nai) / rig, 1] + (1 - nachoice) * 1.4 / rig;

fun[d_] := If[d <= 3, 3 - 2 * (d - 2), 1 + 5 * (d - 3)];
rmax[d_, nf_, rmin_] := rmin * (nf / fun[d]) ^ (1 / Min[d, 3]);
sigma[d_, nf_, phi_, rmin_, y_] :=
  Sqrt[(rmin / rmax[d, nf, rmin]) ^ ((3 - d) / 2) / BesselK[ $\frac{1}{2}(-3 + d)$ ,  $\frac{rmin}{rmax[d, nf, rmin]}$ ]] *
  (2 ^ ((d - 9) / 2) * Gamma[d / 2 - 3 / 2] *
    (1 - (1 + (xp[rmax[d, nf, rmin], phi] * na) ^ 2) ^ (3 / 2 - d / 2)) +
    2 * kt[y, phi] / Sqrt[Pi] * Gamma[d / 2] * xp[rmax[d, nf, rmin], phi] / 2 ^ (3 / 2 - d / 2) /
    (d - 2) * ((1 + 4 * xp[rmax[d, nf, rmin], phi] ^ 2) ^ (1 - d / 2) -
      (1 + xp[rmax[d, nf, rmin], phi] ^ 2 * (4 + na ^ 2)) ^ (1 - d / 2)))];
sigmaexperiment[d_, nf_, phi_, rmin_, t_] := sigma[d, nf, phi, rmin, t] *
  Sqrt[rcoeff[phi]] * sn[phi];

thick = lambda / 1000 * rig / 2 / nai ^ 2;
g = 3.3;
dwm[d_] := 3 * (Log[1 / (1 - (d / 3) ^ g)]) ^ (1 / g);

"Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default"
Text["D_expand=", dexpand, "CVC default=", cvcdefault]
a4 = sigmaexperiment[dexpand, mf, cvcdefault, 1, thick];
b4 = 1 / D[sigmaexperiment[d, mf, cvcdefault, 1, thick], d] /. d -> dexpand
c4 = N[(-a4 + 1 / b4 * dexpand) * b4]
Show[Plot[{sigmaexperiment[d, mf, cvcdefault, 1, thick], 0}, {d, 5 / 3, 4},
  PlotRange -> All], Plot[{1 / b4 * (d - c4)}, {d, 2, 3.5}, PlotRange -> All]]
Plot[{sigmaexperiment[dwm[d], mf, cvcdefault, 1, thick], 0},
  {d, 5 / 3, 3}, PlotRange -> All]

Plot[{sigmaexperiment[dexpand, nf, cvcdefault, 1, thick], 0},
  {nf, 100000, 1000000}, PlotRange -> All]
Plot[{sigmaexperiment[dexpand, mf, phi, 1, thick], 0}, {phi, 0.2, 0.6}, PlotRange -> All]
Plot3D[{sigmaexperiment[d, mf, phi, 1, thick]},
  {d, 5 / 3, 4}, {phi, 0.2, 0.6}, PlotRange -> All, BoxRatios -> {1, 1, 1}]

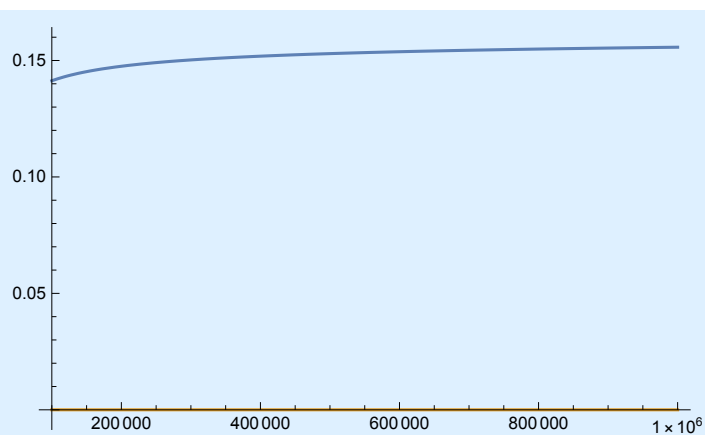
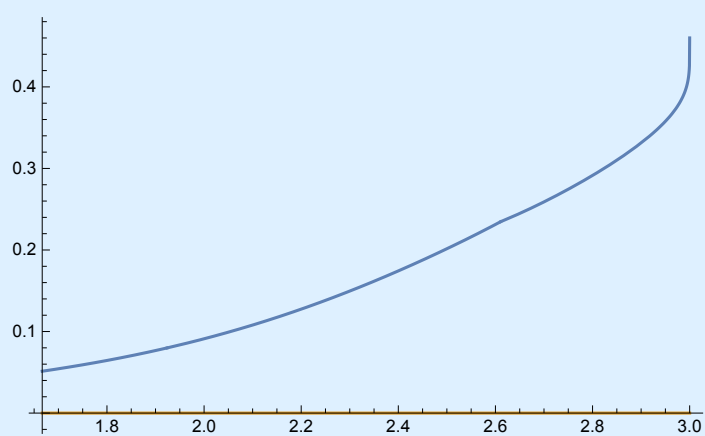
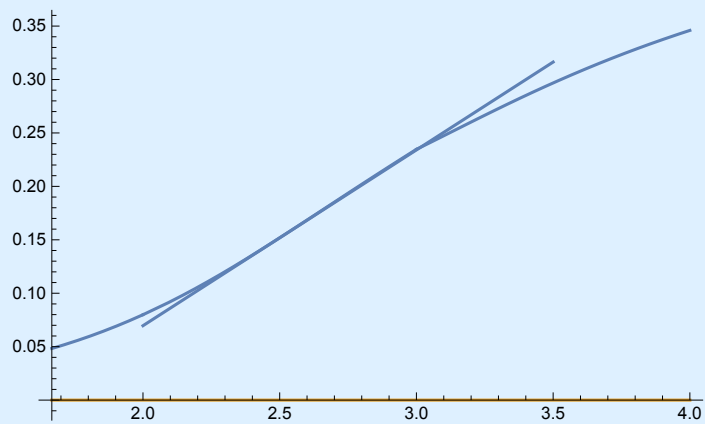
```

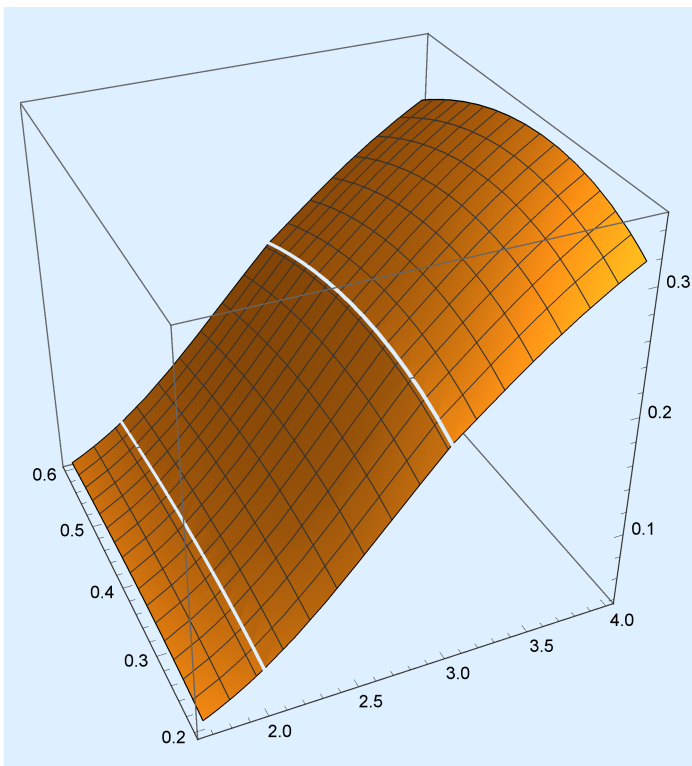
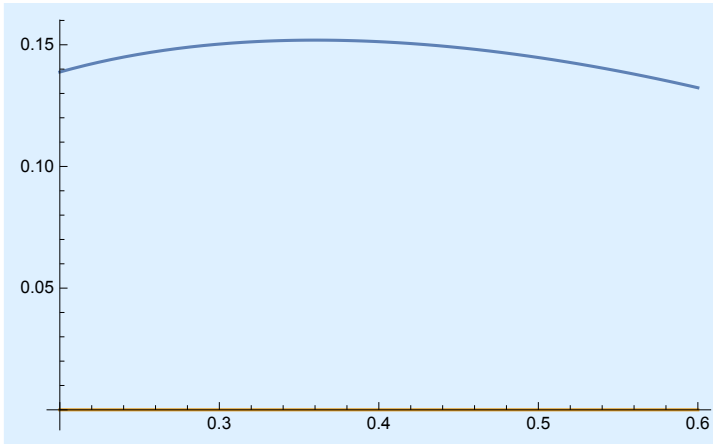
Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default

```
Text[D_expand=, 2.5, CVC default=, 0.35]
```

```
6.07971
```

```
1.57673
```





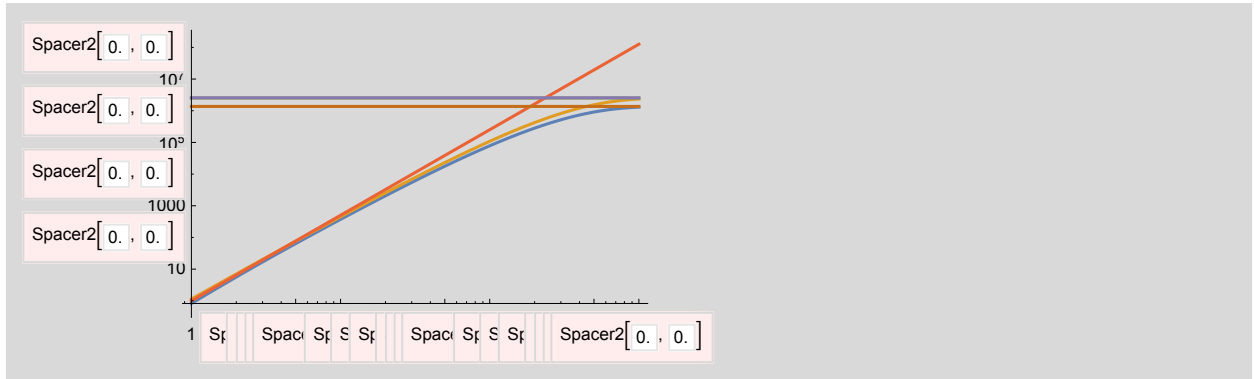
```

acfsz[x_, d_, a_, b_] := (b-3+d ExpIntegralE[-2 + d,  $\frac{x}{b}$ ] - a-3+d ExpIntegralE[-2 + d,  $\frac{x}{a}$ ]) *
  (3 - d) / (1/a^(3 - d) - 1/b^(3 - d));
nfsz[d_, a_, b_] :=  $\frac{6 * (3 - d)}{d} * (b/a)^d * (1 - (a/b)^d) / (1 - (a/b)^(3 - d))$ ;
nsz[r_, d_, a_, b_] :=
  Re[ $\frac{3}{(-a^d b^3 + a^3 b^d) d}$  b3 (-3 + d) * (-2 ad + 2 bd + 2 ad d ( $\frac{r}{a}$ )d Gamma[1 - d,  $\frac{r}{a}$ ] -
    2 bd d ( $\frac{r}{b}$ )d Gamma[1 - d,  $\frac{r}{b}$ ] + ad d ( $\frac{r}{a}$ )d Gamma[2 - d,  $\frac{r}{a}$ ] - bd d ( $\frac{r}{b}$ )d Gamma[2 - d,  $\frac{r}{b}$ ] +
    2 ad d ( $\frac{r}{a}$ )d Gamma[-d,  $\frac{r}{a}$ ] - 2 bd d ( $\frac{r}{b}$ )d Gamma[-d,  $\frac{r}{b}$ ]);

acfwm[r_, d_, rmin_, lc_] :=
  (r/rmin)^( (d - 3) / 2) * BesselK[(d - 3) / 2, r / lc] / BesselK[(d - 3) / 2, rmin / lc];
nwm[r_, d_, rmin_, lc_] :=  $\frac{1}{3 d \text{BesselK}[\frac{1}{2} (-3 + d), \frac{rmin}{lc}]}$  2 $-\frac{1}{2} - \frac{d}{2}$  π r2 ( $\frac{r}{rmin}$ ) $\frac{1}{2} (-1+d)$ 
  rmin Abs[lc] $-\frac{3}{2} - \frac{d}{2}$  Abs[r] $-\frac{3}{2} - \frac{d}{2}$  (2d d Abs[lc]d Abs[r]3 Gamma[ $\frac{1}{2} (-3 + d)$ ]
  HypergeometricPFQ[ $\{\frac{3}{2}\}$ ,  $\{\frac{5}{2}, \frac{5}{2} - \frac{d}{2}\}$ ,  $\frac{r^2}{4 lc^2}$ ] + 24 Abs[lc]3 Abs[r]d Gamma[ $\frac{3}{2} - \frac{d}{2}$ ]
  HypergeometricPFQ[ $\{\frac{d}{2}\}$ ,  $\{-\frac{1}{2} + \frac{d}{2}, 1 + \frac{d}{2}\}$ ,  $\frac{r^2}{4 lc^2}$ ]) / (4 * Pi / 3 * rmin^3);
nfwm[d_, rmin_, lc_] :=  $\frac{2^{(d-3)/2} (lc/rmin)^{\frac{3+d}{2}} 3 * \pi^{1/2} \text{Gamma}[\frac{d}{2}]}{\text{BesselK}[\frac{1}{2} (-3 + d), \frac{rmin}{lc}]}$ ;
nfwmapprox[d_, rmin_, lc_] :=
   $\frac{2^{(d-3)/2} (lc/rmin)^{\frac{3+d}{2}} (0.3 * \text{Abs}[(d-3)]^{1.7+0.21}) 3 * \pi^{1/2} \text{Gamma}[\frac{d}{2}]}{(1.8 - 0.62 * \text{Abs}[d - 3])}$ ;

LogLogPlot[{Re[nsz[x, 2.7, 1, 200]], nwm[x, 2.7, 1, 200], nfwm[2.7, 1, 200], x^2.7,
  nfwmapprox[2.7, 1, 200], nfsz[2.7, 1, 200]}, {x, 1, 1000}, PlotRange -> All]

```

```
Integrate[
  (-a-3+d ExpIntegralE[-2 + d,  $\frac{x}{a}$ ] + b-3+d ExpIntegralE[-2 + d,  $\frac{x}{b}$ ]) * 4 * Pi * x^2, {x, 0, r}]
ConditionalExpression[-  $\frac{8 (a^d - b^d) \pi}{d}$  +
  4 \pi \left( 2 a^d \left( \frac{r}{a} \right)^d \text{Gamma}\left[1 - d, \frac{r}{a}\right] - 2 b^d \left( \frac{r}{b} \right)^d \text{Gamma}\left[1 - d, \frac{r}{b}\right] + a^d \left( \frac{r}{a} \right)^d \text{Gamma}\left[2 - d, \frac{r}{a}\right] -
  b^d \left( \frac{r}{b} \right)^d \text{Gamma}\left[2 - d, \frac{r}{b}\right] + 2 a^d \left( \frac{r}{a} \right)^d \text{Gamma}\left[-d, \frac{r}{a}\right] - 2 b^d \left( \frac{r}{b} \right)^d \text{Gamma}\left[-d, \frac{r}{b}\right] \right), \text{Re}[d] > 0]
```

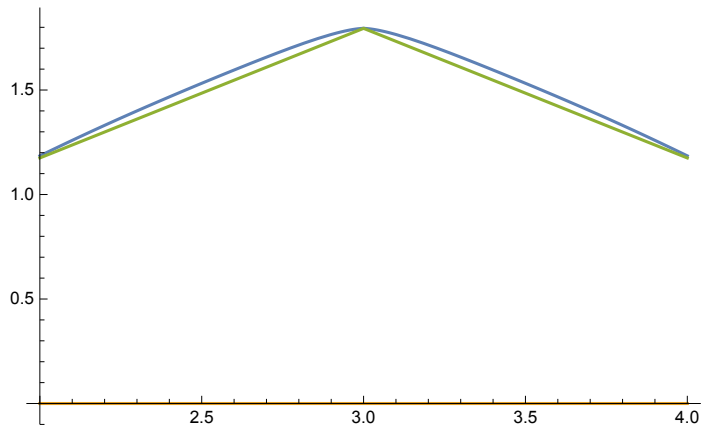
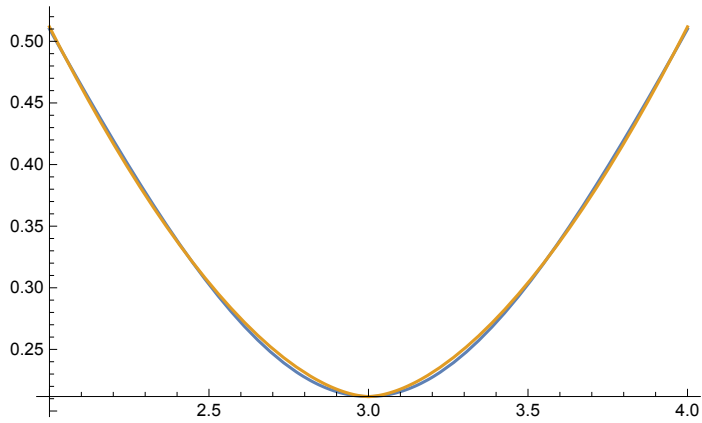
```
Integrate[(x/rmin)^( (d-3)/2) *
  BesselK[(d-3)/2, x/lc]/BesselK[(d-3)/2, rmin/lc] * 4 * Pi * x^2, {x, 0, r}]
ConditionalExpression[
   $\frac{1}{3 d \text{BesselK}\left[\frac{1}{2}(-3+d), \frac{rmin}{lc}\right]} 2^{-\frac{1}{2}-\frac{d}{2}} \pi r^2 \left(\frac{r}{rmin}\right)^{\frac{1}{2}(-1+d)} rmin \text{Abs}[lc]^{-\frac{3}{2}-\frac{d}{2}} \text{Abs}[r]^{-\frac{3}{2}-\frac{d}{2}}$ 
   $\left( 2^d d \text{Abs}[lc]^d \text{Abs}[r]^3 \text{Gamma}\left[\frac{1}{2}(-3+d)\right] \text{HypergeometricPFQ}\left[\left\{\frac{3}{2}\right\}, \left\{\frac{5}{2}, \frac{5}{2} - \frac{d}{2}\right\}, \frac{r^2}{4 lc^2}\right] +
  24 \text{Abs}[lc]^3 \text{Abs}[r]^d \text{Gamma}\left[\frac{3}{2} - \frac{d}{2}\right] \right.$ 
   $\left. \text{HypergeometricPFQ}\left[\left\{\frac{d}{2}\right\}, \left\{-\frac{1}{2} + \frac{d}{2}, 1 + \frac{d}{2}\right\}, \frac{r^2}{4 lc^2}\right] \right), lc r > 0 \&\& \text{Re}[d] > 0]$ 
```

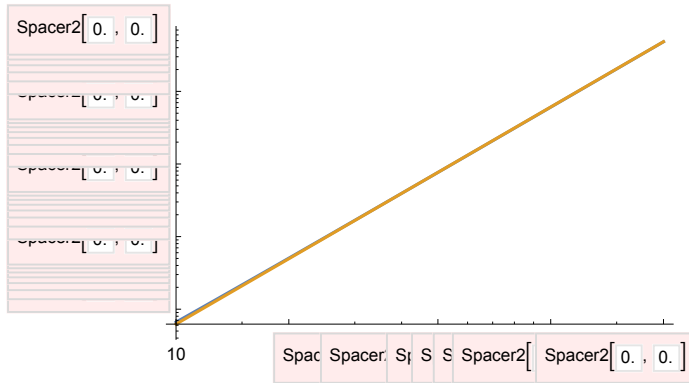
```
Integrate[(x/rmin)^( (d-3)/2) *
  BesselK[(d-3)/2, x/lc]/BesselK[(d-3)/2, rmin/lc] * 4 * Pi * x^2, {x, 0, Infinity}]
ConditionalExpression[
   $\frac{2^{2+\frac{1}{2}(-3+d)} lc^{\frac{3+d}{2}} \pi^{3/2} \left(\frac{1}{rmin}\right)^{\frac{1}{2}(-3+d)} \text{Gamma}\left[\frac{d}{2}\right]}{\text{BesselK}\left[\frac{1}{2}(-3+d), \frac{rmin}{lc}\right]},$ 
   $\text{Re}[d] > 0 \&\& \text{Re}[lc] > 0 \&\& \text{Im}[lc] == 0]$ 
```

```

delta = 0.01;
slp[d_, wx_] :=
  (Log[BesselK[ $\frac{1}{2}(-3+d)$ , 1/(wx+delta)]] - Log[BesselK[ $\frac{1}{2}(-3+d)$ , 1/wx]]) /
  (Log[wx+delta] - Log[wx]);
Plot[{slp[d, 100], (0.3 * Abs[d-3])^1.7 + slp[3, 100]}, {d, 2, 4}]
rat[d_, y_] := BesselK[ $\frac{1}{2}(-3+d)$ , 1/y] / y^(0.3 * Abs[d-3]^1.7 + 0.21);
Plot[{rat[d, 100], 0, rat[3, 100] - 0.62 * Abs[d-3]^1}, {d, 2, 4}]
nfwmapprox[y_, d_] := 
$$\frac{2^{(d-3)/2} y^{\frac{3+d}{2}} (0.3 * \text{Abs}[d-3]^{1.7+0.21}) 3 * \pi^{1/2} \text{Gamma}[\frac{d}{2}]}{(1.8 - 0.62 * \text{Abs}[d-3])}$$
;
nfwm1[y_, d_] := 
$$\frac{2^{(d-3)/2} y^{\frac{3+d}{2}} 3 * \pi^{1/2} \text{Gamma}[\frac{d}{2}]}{\text{BesselK}[\frac{1}{2}(-3+d), 1/y]}$$
;
LogLogPlot[{nfwm1[y, 4], nfwmapprox[y, 4]}, {y, 10, 200}]

```





```

Text["WM ACF"]
surfaceterm = 1;
massfixed = 1;
nachoise = 0;

dexpand = 2.5;
mf = 200000;
cvcdefault = 0.35;
phinucleoplasm = 0.1;

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = nailive;

rig = 1.517 + 1.53 * 0;
rim = 1.337;

riinc = 0.1899 * 0 + (0.1899 * 1 / 2 + 0.17 * 1 / 2 + 0.185 * 0);
purechromdensity = 0.555;
ri[x_] := 1.332 + riinc * purechromdensity * x;
sn[phi_] := Sqrt[phi * (1 - phi)] * (ri[1] - ri[phinucleoplasm]);

ric[phi_] := ri[phi + (1 - phi) * phinucleoplasm];
size[lc_, d_, lmin_, lmax_] :=
  (3 - d) / (1 - (lmin / lmax) ^ (3 - d)) * (lmin / lc) ^ (4 - d) / lmin;
rcoeff[phi_] := (2 * rig / (rig + ric[phi]) * 2 * ric[phi] / (rig + ric[phi]) *
  (rig - ric[phi]) / (rig + ric[phi]) * (rig + rim) ^ 2 / (rig - rim) ^ 2) ^ 2;
kt[l_, phi_] := 2 * Pi / lambda * ric[phi] * l * 1000;
xp[lc_, phi_] := 2 * Pi / lambda * ric[phi] * lc;
na = nachoise * Min[(1.4 + nai) / rig, 1] + (1 - nachoise) * 1.4 / rig;

alpha[d_] := (3 + d) / 2 - (0.3 * Sqrt[(d - 3) ^ 2] ^ 1.7 + 0.21);
beta[d_] := (2^(d-3)/2 * 3 * Pi^(1/2) Gamma[d/2]) / (1.8 - 0.62 * Sqrt[(d - 3) ^ 2]);

```

```

rmax[d_, nf_, rmin_] := rmin * (nf / beta[d]) ^ (1 / alpha[d]);
sigma[d_, nf_, phi_, rmin_, y_] :=
  Sqrt[(rmin / rmax[d, nf, rmin]) ^ ((3 - d) / 2) / BesselK[ $\frac{1}{2}(-3 + d)$ ,  $\frac{rmin}{rmax[d, nf, rmin]}$ ]] *
  (2 ^ ((d - 9) / 2) * Gamma[d / 2 - 3 / 2] *
    (1 - (1 + (xp[rmax[d, nf, rmin], phi] * na) ^ 2) ^ (3 / 2 - d / 2)) * surfaceterm +
    2 * kt[y, phi] / Sqrt[Pi] * Gamma[d / 2] * xp[rmax[d, nf, rmin], phi] / 2 ^ (3 / 2 - d / 2) /
    (d - 2) * ((1 + 4 * xp[rmax[d, nf, rmin], phi] ^ 2) ^ (1 - d / 2) -
      (1 + xp[rmax[d, nf, rmin], phi] ^ 2 * (4 + na ^ 2)) ^ (1 - d / 2)))];
sigmaexperiment[d_, nf_, phi_, rmin_, t_] := sigma[d, nf, phi, rmin, t] *
  Sqrt[rcoeff[phi]] * sn[phi];

thick = lambda / 1000 * rig / 2 / nai ^ 2;
g = 3.3;
dwm[d_] := 3 * (Log[1 / (1 - (d / 3) ^ g)]) ^ (1 / g);

"Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default"
Text["D_expand=", dexpand, "CVC default=", cvcdefault]
a4 = sigmaexperiment[dexpand, mf, cvcdefault, 1, thick];
b4 = 1 / D[sigmaexperiment[d, mf, cvcdefault, 1, thick], d] /. d -> dexpand
c4 = N[(-a4 + 1 / b4 * dexpand) * b4]
Show[Plot[{sigmaexperiment[d, mf, cvcdefault, 1, thick], 0},
  {d, 5 / 3, 4}, PlotRange -> All, AxesLabel -> {Dwm, Sigma}],
  Plot[{1 / b4 * (d - c4)}, {d, 2, 3.5}, PlotRange -> All]]

Text["D_measured_mass_scaling_expand=", dexpand, "CVC default=", cvcdefault]
a5 = sigmaexperiment[dwm[dexpand], mf, cvcdefault, 1, thick];
b5 = 1 / D[sigmaexperiment[dwm[d], mf, cvcdefault, 1, thick], d] /. d -> dexpand
c5 = N[(-a5 + 1 / b5 * dexpand) * b5]
Show[Plot[{sigmaexperiment[dwm[d], mf, cvcdefault, 1, thick], 0},
  {d, 5 / 3, 2.99}, PlotRange -> All, AxesLabel -> {Dmass scaling, Sigma}],
  Plot[{1 / b5 * (d - c5)}, {d, 2, 2.99}, PlotRange -> All]]

Plot[{sigmaexperiment[dexpand, nf, cvcdefault, 1, thick], 0},
  {nf, 100000, 1000000}, PlotRange -> All, AxesLabel -> {Nf, Sigma}]
Plot[{sigmaexperiment[dexpand, mf, phi, 1, thick], 0},
  {phi, 0.2, 0.6}, PlotRange -> All, AxesLabel -> {phi, Sigma}]
Plot3D[{sigmaexperiment[d, mf, phi, 1, thick]}, {d, 5 / 3, 4}, {phi, 0.2, 0.6},
  PlotRange -> All, BoxRatios -> {1, 1, 1}, AxesLabel -> {Dwm, phi, Sigma}]

```

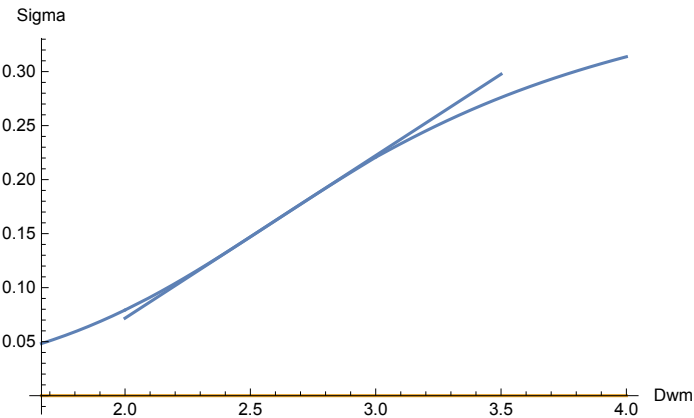
WM ACF

Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default

Text[D_expand=, 2.5, CVC default=, 0.35]

6.64135

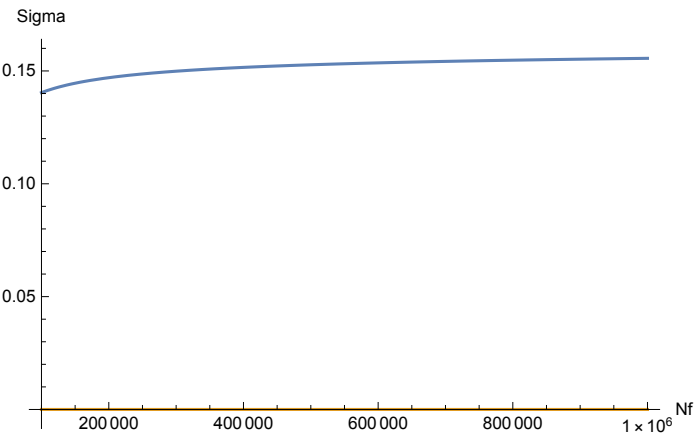
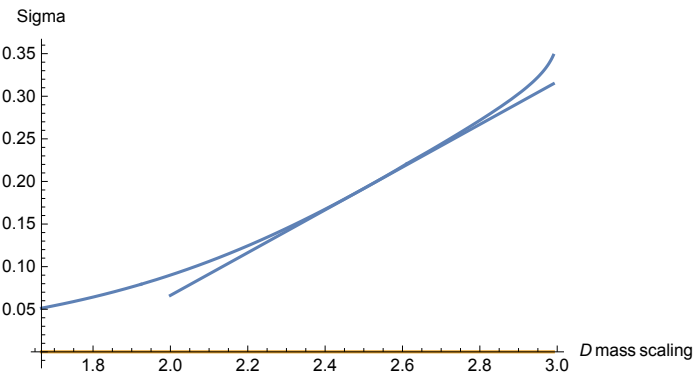
1.52359

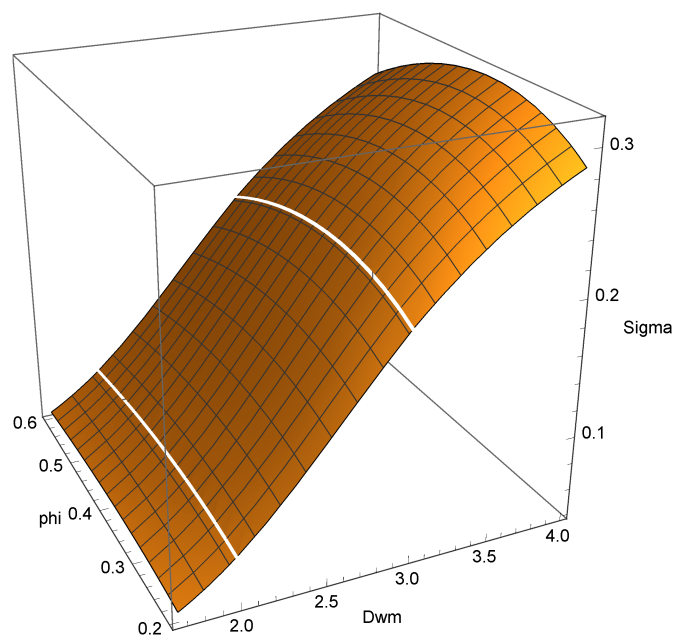
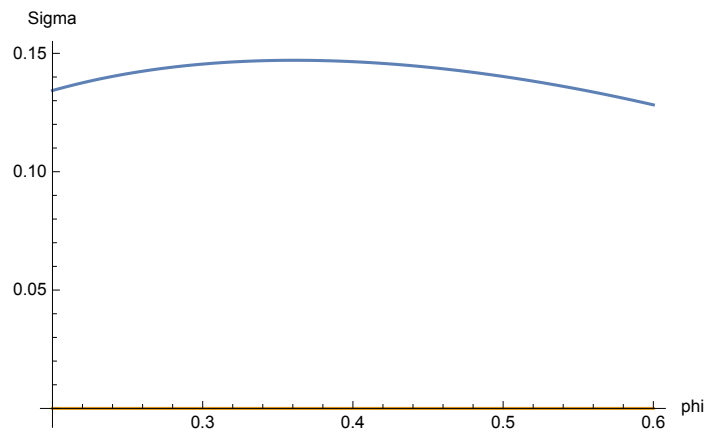


Text [D_measured_mass_scaling_expand=, 2.5, CVC default=, 0.35]

3.98595

1.73599

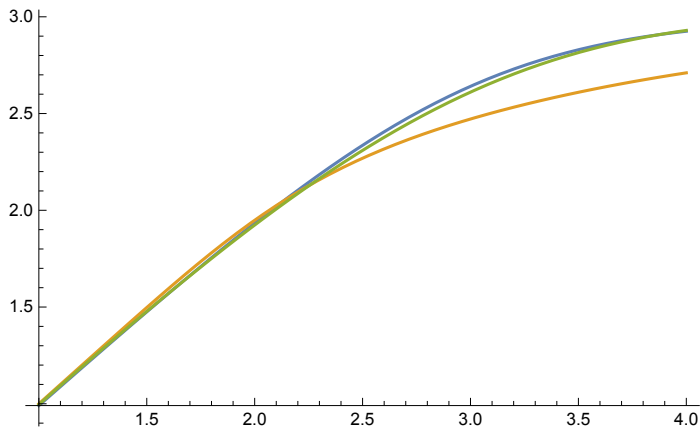




```

delta = 0.01;
rc = 20;
nwmd1[d_, nf_, rmin_] :=
  (Log[nwm[rmax[d, nf, rmin] / 10 + delta, d, rmin, rmax[d, nf, rmin]]] -
   Log[nwm[rmax[d, nf, rmin] / 10, d, rmin, rmax[d, nf, rmin]]]) /
  (Log[rmax[d, nf, rmin] / 10 + delta] - Log[rmax[d, nf, rmin] / 10]);
nwmd2[d_, nf_, rmin_] := (Log[nwm[rc + delta, d, rmin, rmax[d, nf, rmin]]] -
  Log[nwm[rc, d, rmin, rmax[d, nf, rmin]]]) / (Log[rc + delta] - Log[rc]);
mf = 200000;
g = 3.3;
Plot[{nwmd1[d, mf, 1], nwmd2[d, mf, 1], 3 * (1 - Exp[-(d/3)^g])^(1/g)}, {d, 1, 4}]

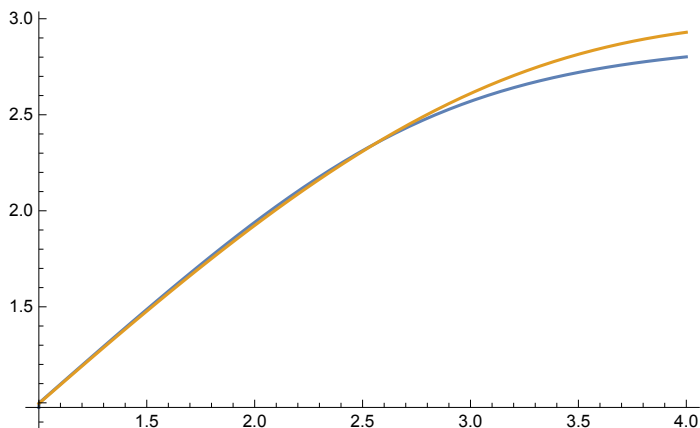
```



```

delta = 0.01;
rc = 20;
nszd1[d_, nf_, rmin_] :=
  (Log[nsz[rmax[d, nf, rmin] / 10 + delta, d, rmin, rmax[d, nf, rmin]]] -
   Log[nsz[rmax[d, nf, rmin] / 10, d, rmin, rmax[d, nf, rmin]]]) /
  (Log[rmax[d, nf, rmin] / 10 + delta] - Log[rmax[d, nf, rmin] / 10]);
nszd2[d_, nf_, rmin_] := (Log[nsz[rc + delta, d, rmin, rmax[d, nf, rmin]]] -
  Log[nsz[rc, d, rmin, rmax[d, nf, rmin]]]) / (Log[rc + delta] - Log[rc]);
mf = 200000;
g = 3.3;
Plot[{nszd1[d, mf, 1], 3 * (1 - Exp[-(d/3)^g])^(1/g)}, {d, 1, 4}]

```



```

Text["Size Distribution method, an alternate method to calculate Nf(lc)"]
surfaceterm = 1;

```

```

massfixed = 1;
nachoise = 0;

dexpand = 2.5;
mf = 200000;
cvcdefault = 0.35;
phinucleoplasma = 0.1;

lambda = 583;
naistorm = 0.4;
nailive = 0.55;
nai = nailive;

rig = 1.517 + 1.53 * 0;
rim = 1.337;

riinc = 0.1899 * 0 + (0.1899 * 1 / 2 + 0.17 * 1 / 2 + 0.185 * 0);
purechromdensity = 0.555;
ri[x_] := 1.332 + riinc * purechromdensity * x;
sn[phi_] := Sqrt[phi * (1 - phi)] * (ri[1] - ri[phinucleoplasma]);

ric[phi_] := ri[phi + (1 - phi) * phinucleoplasma];
size[lc_, d_, lmin_, lmax_] :=
  (3 - d) / (1 - (lmin / lmax)^(3 - d)) * (lmin / lc)^(4 - d) / lmin;
rcoeff[phi_] := (2 * rig / (rig + ric[phi]) * 2 * ric[phi] / (rig + ric[phi]) *
  (rig - ric[phi]) / (rig + ric[phi]) * (rig + rim)^2 / (rig - rim)^2)^2;
kt[l_, phi_] := 2 * Pi / lambda * ric[phi] * 1 * 1000;
xp[lc_, phi_] := 2 * Pi / lambda * ric[phi] * lc;
na = nachoise * Min[(1.4 + nai) / rig, 1] + (1 - nachoise) * 1.4 / rig;

rmax[d_, y_, rmin_] :=
  NSolve[ $\frac{6 * (3 - d)}{d} * b^d / (1 - (1 / b)^(3 - d)) = y, b, Reals][[2, 1, 2]];
s2[lc_, y_, phi_] := 2 / Pi * (kt[y, phi] * xp[lc, phi]^3 * na^2) /
  (1 + xp[lc, phi]^2 * (4 + na^2)) / (1 + xp[lc, phi]^2 * 4) +
  surfaceterm * 1 / 4 * (1 - 1 / Sqrt[1 + (xp[lc, phi] * na)^2]);
sigma[d_, lmin_, lmax_, y_, phi_] := Sqrt[NIntegrate[(1 / lc)^(4 - d) *
  s2[lc, y, phi] * (3 - d) / (1 / lmin^(3 - d) - 1 / lmax^(3 - d)), {lc, lmin, lmax}]];
sigmaexperiment[d_, nf_, phi_, rmin_, t_] :=
  sigma[d, rmin, rmax[d, nf, rmin], t, phi] * Sqrt[rcoeff[phi]] * sn[phi];

sigmapl[d_, lmin_, y_, phi_] := Sqrt[NIntegrate[
   $\frac{(3 - d)^2 * \Gamma[3 - d] * (lmin / lc)^{4 - d}}{\Gamma[4 - d] * lmin}$  * s2[lc, y, phi], {lc, lmin, Infinity}]];
sigmaexperimentpl[d_, phi_, rmin_, t_] :=
  sigmapl[d, rmin, t, phi] * Sqrt[rcoeff[phi]] * sn[phi];

thick = lambda / 1000 * rig / 2 / nai^2;
g = 3.3;$ 
```



```

dwm[d_] := 3 * (Log[1 / (1 - (d/3)^g)])^(1/g);

"Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default"
Text["D_expand=", dexpand, "CVC default=", cvcdefault]
a4 = sigmaexperiment[dexpand, mf, cvcdefault, 1, thick];
b4 = 1/D[sigmaexperiment[d, mf, cvcdefault, 1, thick], d] /. d -> dexpand
c4 = N[(-a4 + 1/b4 * dexpand) * b4]
Show[Plot[{sigmaexperiment[d, mf, cvcdefault, 1, thick], 0},
  {d, 5/3, 4}, PlotRange -> All, AxesLabel -> {Dsd, Sigma}],
  Plot[{1/b4 * (d - c4)}, {d, 2, 3.5}, PlotRange -> All]]

Text["D_measured_mass_scaling_expand=", dexpand, "CVC default=", cvcdefault]
a5 = sigmaexperiment[dwm[dexpand], mf, cvcdefault, 1, thick];
b5 = 1/D[sigmaexperiment[dwm[d], mf, cvcdefault, 1, thick], d] /. d -> dexpand
c5 = N[(-a5 + 1/b5 * dexpand) * b5]
Show[Plot[{sigmaexperiment[dwm[d], mf, cvcdefault, 1, thick], 0},
  {d, 5/3, 2.99}, PlotRange -> All, AxesLabel -> {D mass scaling, Sigma}],
  Plot[{1/b5 * (d - c5)}, {d, 2, 2.99}, PlotRange -> All]]

Plot[{sigmaexperiment[dexpand, nf, cvcdefault, 1, thick], 0},
  {nf, 100000, 1000000}, PlotRange -> All, AxesLabel -> {Nf, Sigma}]
Plot[{sigmaexperiment[dexpand, mf, phi, 1, thick], 0},
  {phi, 0.2, 0.6}, PlotRange -> All, AxesLabel -> {phi, Sigma}]
Show[Plot[{sigmaexperiment[d, mf, cvcdefault, 1, thick], 0}, {d, 5/3, 4}, PlotRange -> All,
  AxesLabel -> {Dsd, Sigma}], Plot[{sigmaexperimentpl[d, cvcdefault, 1, thick],
  sigmaexperiment[dwm[d], mf, cvcdefault, 1, thick],
  sigmaexperiment[dwm[d], mf, cvcdefault, 1, thick]}, {d, 5/3, 2.98}, PlotRange -> All]]
Plot3D[{sigmaexperiment[d, mf, phi, 1, thick]}, {d, 5/3, 4}, {phi, 0.2, 0.6},
  PlotRange -> All, BoxRatios -> {1, 1, 1}, AxesLabel -> {Dsd, phi, Sigma}]

```

Size Distribution method, an alternate method to calculate Nf(lc)

Linear approximation for Sigma_experiment at D=D_expand and phi-CVC_default

Text[D_expand=, 2.5, CVC default=, 0.35]

NSolve: This system cannot be solved with the methods available to NSolve.

Part: Part specification NSolve[$\frac{6 b^d (3 - d)}{(1 - \text{Power}[\ll 2 \gg]) d} == 200000, b,$][2, 1, 2] is longer than depth of object.

Part: Part specification NSolve[$\frac{6 b^d (3 - 1. d)}{(1 - 1. \text{Power}[\ll 2 \gg]) d} == 200000, b,$][2, 1, 2] is longer than depth of object.

Part: Part specification NSolve[$\frac{6 b^d (3 - 1. d)}{(1 - 1. \text{Power}[\ll 2 \gg]) d} == 200000, b,$][2, 1, 2] is longer than depth of object.

General: Further output of Part::partd will be suppressed during this calculation.

NIntegrate: lc = NSolve[$\frac{6 b^d (3 - 1. d)}{(1 - 1. \text{Power}[\ll 2 \gg]) d} == 200000, b,$][2, 1, 2] is not a valid limit of integration.

NIntegrate: $lc = \text{NSolve}\left[\frac{6. b^d (3. - 1. d)}{(1. - 1. \text{Power}[\ll 2 \gg]) d} == 200000., b, \text{ } \diamond : \diamond\right][[2, 1, 2]]$ is not a valid limit of integration.

NIntegrate: $lc = \text{NSolve}\left[\frac{6. b^d (3. - 1. d)}{(1. - 1. \text{Power}[\ll 2 \gg]) d} == 200000., b, \text{ } \diamond : \diamond\right][[2, 1, 2]]$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

NSolve: Warning: #3 is not a valid domain specification. Assuming it is a variable to eliminate.

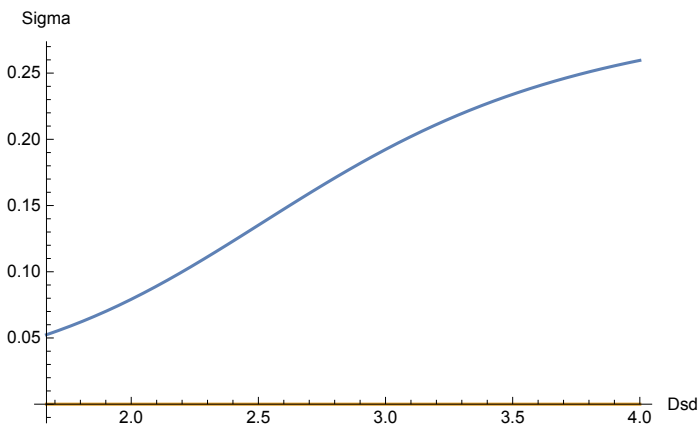
NSolve: Warning: #3 is not a valid domain specification. Assuming it is a variable to eliminate.

NSolve: Warning: #3 is not a valid domain specification. Assuming it is a variable to eliminate.

General: Further output of NSolve::bdomv will be suppressed during this calculation.

{ }

{ }



Text [D_measured_mass_scaling_expand=, 2.5, CVC default=, 0.35]

Part: Part specification $\text{NSolve}\left[\frac{2 b^3 \text{Log}[\ll 1 \gg]^{0.30303} (3. - 3. \text{Log}[\ll 1 \gg])^{0.30303}}{(1. - \text{Power}[\ll 2 \gg]) \text{Log}\left[\frac{1}{\text{Plus}[\ll 2 \gg]}\right]^{0.30303}} == 200000., b, \text{ } \diamond : \diamond\right][[2, 1, 2]]$ is longer than depth of object.

Part: Part specification $\text{NSolve}\left[\frac{2. b^3. \text{Log}[\ll 1 \gg]^{0.30303} (3. - 3. \text{Log}[\ll 1 \gg])^{0.30303}}{(1. - 1. \text{Power}[\ll 2 \gg]) \text{Log}\left[\frac{1}{\text{Plus}[\ll 2 \gg]}\right]^{0.30303}} == 200000., b, \text{ } \diamond : \diamond\right][[2, 1, 2]]$ is longer than depth of object.

Part: Part specification $\text{NSolve}\left[\frac{2. b^3. \text{Log}[\ll 1 \gg]^{0.30303} (3. - 3. \text{Log}[\ll 1 \gg])^{0.30303}}{(1. - 1. \text{Power}[\ll 2 \gg]) \text{Log}\left[\frac{1}{\text{Plus}[\ll 2 \gg]}\right]^{0.30303}} == 200000., b, \text{ } \diamond : \diamond\right][[2, 1, 2]]$ is longer than depth of object.

General: Further output of Part::partd will be suppressed during this calculation.

NIntegrate: $lc = \text{NSolve}\left[\frac{2. b^3. \text{Log}[\ll 1 \gg]^{0.30303} (3. - 3. \text{Log}[\ll 1 \gg])^{0.30303}}{(1. - 1. \text{Power}[\ll 2 \gg]) \text{Log}\left[\frac{1}{\text{Plus}[\ll 2 \gg]}\right]^{0.30303}} == 200000., b, \text{ } \diamond : \diamond\right][[2, 1, 2]]$ is not a valid limit of integration.

NIntegrate: $lc = \text{NSolve}\left[\frac{2. b^3. \text{Log}[\ll 1 \gg]^{0.30303} (3. - 3. \text{Log}[\ll 1 \gg]^{0.30303})}{(1. - 1. \text{Power}[\ll 2 \gg]) \text{Log}\left[\frac{1}{\text{Plus}[\ll 2 \gg]}\right]^{0.30303}} == 200000., b, \text{ } \right] \ll 2, 1, 2 \gg$ is not a valid limit of integration.

NIntegrate: $lc = \text{NSolve}\left[\frac{2. b^3. \text{Log}[\ll 1 \gg]^{0.30303} (3. - 3. \text{Log}[\ll 1 \gg]^{0.30303})}{(1. - 1. \text{Power}[\ll 2 \gg]) \text{Log}\left[\frac{1}{\text{Plus}[\ll 2 \gg]}\right]^{0.30303}} == 200000., b, \text{ } \right] \ll 2, 1, 2 \gg$ is not a valid limit of integration.

General: Further output of NIntegrate::nlim will be suppressed during this calculation.

NSolve: Warning: #3 is not a valid domain specification. Assuming it is a variable to eliminate.

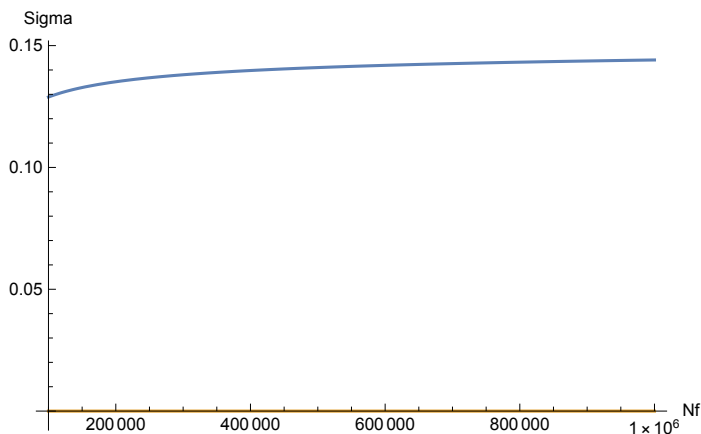
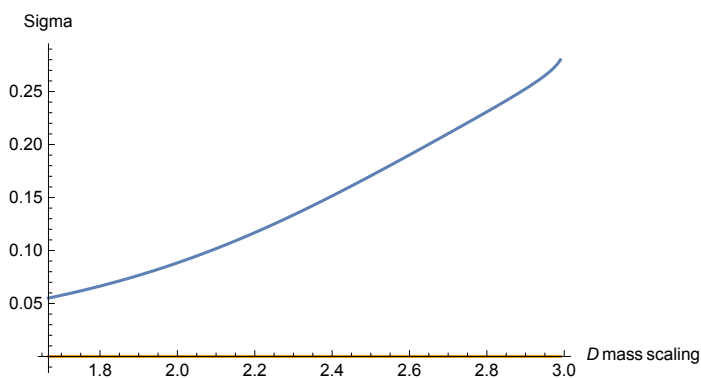
NSolve: Warning: #3 is not a valid domain specification. Assuming it is a variable to eliminate.

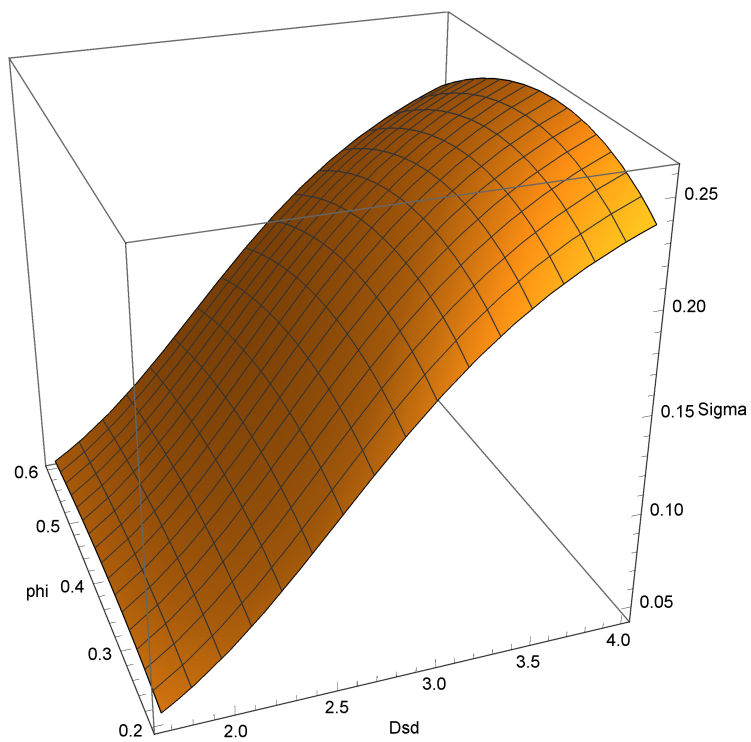
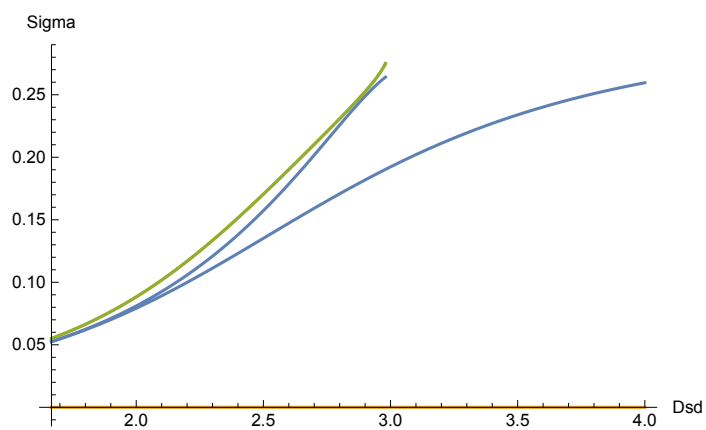
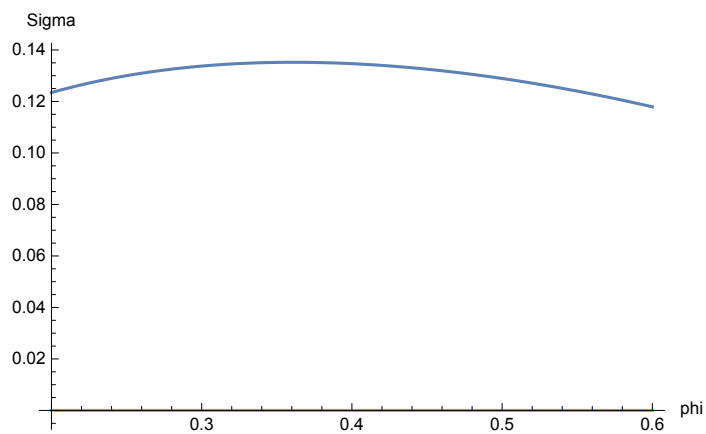
NSolve: Warning: #3 is not a valid domain specification. Assuming it is a variable to eliminate.

General: Further output of NSolve::bdomv will be suppressed during this calculation.

{ }

{ }





```

Integrate[ $\frac{(3-d)^2 \text{Gamma}[3-d] * (\text{lmin}/\text{lc})^{4-d}}{\text{Gamma}[4-d] * \text{lmin}}$  * Exp[-r/lc], {lc, lmin, Infinity}]

acfp1[r_, d_, lmin_] :=  $\frac{(3-d) \text{lmin}^3 \left( \text{lmin}^{-d} r^d \text{Gamma}[3-d] - \left( \frac{\text{lmin}}{r} \right)^{-d} \text{Gamma}\left[3-d, \frac{r}{\text{lmin}}\right] \right)}{r^3}$ ;

LogLogPlot[{acfp1[r, 2.5, 1]}, {r, 0.2, 2000}]

ConditionalExpression[- $\frac{(-3+d) \text{lmin}^3 \left( \text{lmin}^{-d} r^d \text{Gamma}[3-d] - \left( \frac{\text{lmin}}{r} \right)^{-d} \text{Gamma}\left[3-d, \frac{r}{\text{lmin}}\right] \right)}{r^3}$ ,
  Re[lmin] ≥ 0 && ((Re[d] ≤ 2 && Re[r] > 0) || (2 < Re[d] < 3 && Re[r] ≥ 0))

```

